Three Detectives in a Room: Investigating Character Consistency in Multi-Agent LLM Collaboration

Yifei Chen

March 12, 2026

ACKNOWLEDGEMENT

I thank my instructor Ms. Polina Tsvilodub for encouraging and guiding me on the completion of this thesis.

ABSTRACT

This thesis investigates the challenge of character consistency in multi-agent large language model (LLM) collaboration. While LLMs are increasingly used for role-playing, they often struggle to maintain a specific character over long interactions. This study uses a simulation-based design where three agents, role-playing as Sherlock Holmes, Hercule Poirot, and Miss Marple, work together to solve murder mystery cases. The research evaluates character consistency across three linguistic levels: lexical, syntactic, and discourse. A fine-tuned BERT-based classifier is also used as a stylistic benchmark to measure how well the agents stay in character. The results show that collaborative pressure significantly reduces character consistency. While agents partially preserve their characters, their linguistic styles often converge toward a generic, neutral "AI register" as the dialogue progresses. The study finds that syntactic depth is the most reliable indicator for capturing shifts in character consistency. Furthermore, clustering analysis reveals that while detectives are distinct in literary texts, they become representationaly similar during collaboration, a phenomenon termed character contamination. This thesis provides a new multi-level framework for measuring how social dynamics affect the character consistency of LLM agents. Code and data are available on GitHub. [1]

[1] https://github.com/devychen/detective_sim

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1 Introduction

1.1 Application Background

Large language models (LLMs) have achieved remarkable success on a wide range of NLP tasks (text generation, summarisation, instruction following, etc.) [Ji et al., 2025, Guo et al., 2024] Beyond these general applications, there is growing interest in using LLMs as interactive role-playing agents in simulation environments, entertainment, education, and social settings. For example, LLMs are now being deployed as 'stand-ins' for students, patients, voters or citizens in educational tutoring, therapy, social simulation [Park et al., 2022, 2023] and behavioural studies [Aher et al., 2023], and also in entertainments such as games [Xu et al., 2024b]. In such applications, each agent is typically given a persona or fictional character profile that it should be consistently embody. The profile contains role-playing background information such as career, backstory, personality traits, etc. This ability of LLM agents to generate realistic multi-agent dialogues, simulate psychological experiments or drive richly interactive games, opens up novel possibilities at scale and with reproducibility that would be impossible with human actors. For instance, recent work has shown LLMs being used to generate realistic populations in multi-agent social simulations, and to engage in cooperative [Guo et al., 2026] or adversarial [Baltaji et al., 2024] group tasks.

However, a major challenge in these setting is character consistency: an LLM agent must contain its assigned persona throughout a dialogue or collaborative task. In practice, off-the-shelf LLMs often drift from their personas, contradict earlier statements or display inconsistent traits, especially over longer interactions [Choi et al., 2024, Bhandari et al., 2025, Abdulhai et al., 2025]. As Ji et al. [2025] note, current LLMs "lack [. . .] fine-grained role awareness," which limits their ability to provide truly personalised, role-consistent interactions. In collaborative multi-agent scenarios, this issue is amplified: agents may influence one another and inadvertently dilute distinct personality features. Maintaining credible character behaviour is thus essential for user satisfaction and system validity in these applications.

1.2 Character Consistency In Linguistic Context

This study approach character consistency from a linguistic perspective. in sociolinguistics and stylometry, a speaker's or a character's identity is not viewed as a fixed label but as continuously construction through language use [Przystalski et al., 2025]. In other words, how a character speaks conveys their persona, including choice of words, syntax, discourse style. For example, stylometric studies of television dialogue show that fictional characters develop distinct idiolects: Sheldon Cooper in the show The Big Bang Theory habitually uses formal, scientific vocabulary and passive constructions, whereas Penny's speech is more colloquial and conversational [Van Zyl and Botha, 2016]. Language therefore serves as a window into attitude, beliefs, and personality.

Following this view, this study posits that the character consistency of a LLM-based role-playing agent could be manifested in the linguistic profile of its outputs. If an agent truly embodies a persona, its generated responses should consistently reflect that persona's linguistic style. In this study, character consistency is therefore treated as fundamentally a linguistic style problem: this study expects to detect (in)consistency by analysing lexical, syntactic and discourse-level patterns in the generated dialogue text. This aligns with sociolinguistic theory that identity is performed and negotiated through language [Bucholtz and Hall, 2005]. By focusing on linguistic features, consistency could be quantified in a principled way.

1.3 Research Gap

Despite the importance of character consistency, existing work on LLM-based agents has largely focused on model design, training paradigm, or learning methods, rather than systematic evaluation. And even evaluation rarely was designed from linguistic perspective. Much prior research has tried to imbue LLMs with persona via instruction fine-tuning such as the character-specific LLMs in CharacterGLM [Zhou et al., 2023] or the Ditto [Li et al., 2021, Lu et al., 2024], chain-of-thought [Chae et al., 2023] or contrastive Reinforcement Learning fine-tuning [Shea and Yu, 2023]. Moreover, they often use coarse or costly evaluation methods: performance is measured by high-level metrics of role-fidelity such as Big Five, Myers-Biggs Type Indicator (MBTI), or human judgements, rather than by analysing the language itself.

1.4 Research Questions

This thesis investigated character consistency in LLM-based agents through the following questions:

1. **RQ1**: To what extent do LLMs maintain character consistency during collaborative reasoning tasks?

   Hypothesis: LLMs will partially preserve assigned personas but will exhibit character inconsistency over longer multi-turn dialogues.

2. **RQ2**: How can multi-level linguistic metrics capture character consistency?

   Hypothesis: Combining lexical, syntactic, and discourse features will better reflect persona fidelity than any single level alone.

3. **RQ3**: Which metrics are most reliable capture it?

   Hypothesis: Metrics that correlate strongly with a dedicated persona classifier's confidence scores (e.g. classification probability that an utterance matches the persona) will be more predictive of true consistency

These questions guide a comprehensive evaluation of character consistency, from empirical measurement to alignment with predictive models.

1.5 Contribution

In addressing the questions above, this thesis makes the following contributions:

1. **A linguistically grounded multi-level evaluation framework**: This study proposes a systematic evaluation framework that combines lexical, syntactic and discourse-level metrics to assess character consistency in multi-agent dialogue.

2. **Classifier Validation and Calibration**: This study introduces a methodology to validate character consistency metrics by correlating them with a character classification model's outputs. By treating consistency scores as pseudo-probabilities, this study examines their calibration against classifier confidence, identifying which metrics are most predictive and reliable.

Together, these contributions aim to fill the identified gap: providing a linguistically grounded multi-dimensional evaluation of character consistency in LLM-based agents, especially under collaborative conditions. All components are designed to be extensible for future research in role-playing driven conversational AI.

2 Background And Related Work

2.1 Character As A Linguistic Construct

From the perspective of discourse and sociolinguistic theory, an individual's identity (in this scenario, a character's persona) is continuously enacted through language. In sociolinguistics, language is understood to construct identity: speakers adopt styles and linguistic cues that associate them with social groups stereotypes or affective states. Similarly, in narrative, Characterisation is manifested by the way character speak. For example, Culpeper [2001] argued that "an individual's feelings, behaviours, and cognitive states are represented by and through language", thus language use serves as a primary source of information about a character. In other words, characters are written into existence by their linguistic choices.

This view is borne out by stylometric analyses. Stylometry is the quantitative study of linguistic style, originally developed for authorship attribution. Such methods have been applied to fictional dialogues to distinguish characters. For instance, a stylometric analysis of The Big Bang Theory dialogues [Van Zyl and Botha, 2016] showed the character Sheldon Cooper consistently differs from other characters in vocabulary and grammar: he uses more formal, scientific terms and favours passive voice, giving his language an "expository" quality distinct from his friends' colloquial style. These distinct styles emerged despite being written by multiple script authors. Similarly, stylometric analysis of Jane Austen novels by Burrows [1987] and others have demonstrated that characters exhibit quantifiable differences in function word usage and lexical patterns. These examples illustrate that character identity leaves a detectable imprint on language.

Modern stylometry research whether focusing on fictional characters or not reinforces this idea. For authorship attribution or model identification, features such as character n-gram, word frequencies, syntactic structure and sentiments serve as "linguistic fingerprints" of an author or text generator. Studies show that a combination of lexical and grammatical features can distinguish texts written by different LLMs or humans with high accuracy [Przystalski et al., 2025]. They note that LLM-generated text tends to exhibit greater grammatical standardisation and systematic usage patterns compared to human text. These stylistic markers consistently reflect the underlying features of the speaker or model, although maybe subtle.

In sum, the background literature suggests that language style is a valid proxy for character consistency. If an LLM is playing a consistent character, its utterances should collectively form a coherent style profile. Conversely, a sudden in language choices might signal a break in character. This perspective motivates current approach: this study will use linguistic style metrics as indirect measures of character consistency.

In this thesis, the term persona and character is distinguished. This study uses persona to refer to the explicitly assigned profile or identity specification provided to the model through prompting. In contrast, character refers to the identity as realised through linguistic behaviour in dialogue. While persona concerns the intended role, character concerns its performed and observable manifestation. Character consistency therefore denotes the stability of this linguistic realisation across turns.

2.2 Character Modelling In LLMs

In computational linguistics, character modelling typically means conditioning a language model on a character description or background. The simplest method is prompt-based conditioning, where a textual character profile (e.g. "You are Sherlock Holmes, a consulting detective who ...") is prepended to the conversation prompt. In this way, even a genera-purpose LLM can be guided to role-play that character. Recent studies confirm that prompting with explicit role profile can effectively steer LLMs. For example, Chen et al. [2023a], Tu et al. [2024] show that chat models given a well-crafted character prompt can emulate that persona in dialogue. This in-context approach leverages the LLM's existing knowledge without fine-tuning, and can produce consistent character-specific responses with minimal examples.

More advanced methods incorporate character information into model architecture or training. Early works without LLMs (with language models for example) fused memory networks or personal databases into dialogue models [Zhang et al., 2018]. With LLMs, researchers have built character-specific LLMs by further training on persona data. For example, CharacterGLM [Zhou et al., 2023] and Ditto [Li et al., 2021, Lu et al., 2024] are instruction-tuned models with enhanced persona consistency for specific characters. Reinforcement learning techniques have also been applied. For example, Shea and Yu [2023] use RLHF to better align models with persona traits, and Chen et al. [2025] generate preference data with GPT-4 followed by Direct Preference Optimisation (DPO) to refine LLM persona behaviour. These engineering solutions can improve consistency, but they often require expensive data collection or fine-tuning.

Despite these advances, a common observation is that character inconsistency remains a problem. For example, Choi et al. [2024] show that even when LLMs are primed with a persona, they may gradually shift away from it during a long conversation. In their study of nine LLMs, larger models tended to experience greater identity , and simply specifying a persona did not fully prevent its degradation. Similarly, [Frisch and Giulianelli, 2024] found that persona prompts influence language style only up to a point and over multiple turns the models' output can change qualitatively. This suggests that while persona conditioning methods exist, they do not guarantee indefinite consistency.

In sum, character and persona modelling in LLMs spans a range of techniques from prompt engineering to specialised training. But issue of inconsistency and erosion persist. Many prior works focus on how to inject persona and prevent inconsistency, whereas this study focus on assessing how well the character is maintained in the generated text.

2.3 Evaluation Of Character Consistency

2.3.1 Human Evaluation

Previous work have used a number of approaches. Human evaluation is the most traditional approach and is often regarded as the most reliable form of assessment. Researchers typically recruit human annotators, who may include domain experts or fans of the character being simulated. These evaluators assess the agent's performance across multiple dimensions. This can provide nuanced feedback, but is costly and subjective.

Common criteria include consistency, which measures whether the agent's responses align with the predefined character profile, human-likeness, and engagement, which reflects how interesting or immersive the interaction appears to users. For example, Yang et al. [2024], Zhou et al. [2023] report human evaluations of their character customised model CharacterGLM and SimsChat agent for character consistency, and Tu et al. [2024] use human annotators on their CharacterEval benchmark. Zhu et al. [2024] distributed a survey to human participants to capture their gameplay experience and satisfaction levels with agents.

Two common formats are used. In pointwise evaluation, annotators rate the output of a single model according to predefined criteria [Zhou et al., 2023]. In pairwise comparison, annotators compare responses produced by two models and determine which performs better. For example, in role-playing detective games [Zhu et al., 2024], human participants may interact directly with an agent and then evaluate its performance using criteria such as narrative progression or character immersion.

2.3.2 LLM-as-a-judge

Because human evaluation is costly and difficult to reproduce, recent studies increasingly employ large language models as automated evaluators. In this approach, a stronger model such as GPT-4 is used to assess the responses produced by other agents [Chen et al., 2023b, 2024b, Shao et al., 2023].

Researchers may design structured interview procedures in which the evaluation model asks the agent a series of questions. The responses are then assessed along predefined dimensions, such as memory consistency, value alignment, personality traits, hallucination tendency, and behavioural stability [Shao et al., 2023, Park et al., 2023]. The evaluation model can also rank responses generated by multiple agents according to specified criteria, such as their compatibility with the scenario, character attributes, or interpersonal relationships [Chen et al., 2023b, 2024b].

Borrowing from psychology, some studies apply structured personality tests. Jiang et al. [2024] and Sorokovikova et al. [2024] feed LLM outputs into the Big Five Inventory to score train alignment. Pan and Zeng [2023] use the Myers-Biggs Type Indicator (MBTI) to categorise LLM "personalities" and observe that each model tends towards a stable MBTI type. Wang et al. [2024c] perform interview-style tests for personality fidelity. These methods can quantify high-level personality consistency, but they rely on the assumption that LLM output can be meaningfully mapped to human trait scales which is still an open question. As for fine-grained error analysis, LLM-based evaluators can further identify specific types of failure. These include out-of-character behaviour (OOC), internal contradictions, and repetitive responses [Zhou et al., 2023].

2.3.3 Reference-based Automatic Metrics

Another line of evaluation relies on existing textual resources or reference answers. One popular dimension is the **text overlaps**. Metrics such as BLEU and ROUGE-L are commonly used to measure lexical overlap between the generated responses and reference texts. In role-playing settings, these references may include original scripts, character dialogue, or curated conversation datasets [Chen et al., 2023b, Wang et al., 2024a,b].

Another dimension is the **semantic similarity** implemented by embedding-based approaches. By computing cosine similarity between generated responses and character descriptions, researchers can evaluate whether the agent's output remains consistent with the background knowledge or identity profile of the character [Wang et al., 2024b].

2.3.4 Psychological And Behavioural Metrics

In addition to above directions, recent work has begun to evaluate LLM agents from a psychological perspective. One is **psychological scale testing**. Frameworks such as Wang et al. [2024c] assess personality fidelity by asking agents questions derived from established psychological inventories. Examples include the Big Five Inventory (BFI) or other personality assessment tools [Pan and Zeng, 2023]. The resulting scores are used to determine whether the agent's responses align with the expected personality traits of the target character.

Another aspect is the **behaviour and motivation prediction**. In narrative or scenario-based environments, agents may also be evaluated based on their ability to predict or simulate the actions of a character. For instance, in detective scenarios, researchers examine whether the agent can correctly anticipate a character's next move or infer hidden motivations [Shao et al., 2023, Chen et al., 2024a, Wang et al., 2024c, Xu et al., 2024a].

2.3.5 Task-specific Performance Evaluation

When agents operate within collaborative or problem-solving environments, task performance becomes another important evaluation dimension. One metric is the **task accuracy**. In the scenarios similar to this study, that is, investigative games or detective simulations, agents may be evaluated according to their ability to collect relevant clues, identify the culprit, and perform logical reasoning [Wu et al., 2024].

Another aspect is the **team efficiency**. In multi-agent systems, evaluation may also consider group-level outcomes, such as task completion time, communication efficiency, and leadership dynamics within the team [Guo et al., 2026].

2.4 Linguistic Metrics

The methods described above provide useful perspectives on role-playing agents, yet they remain largely fragmented. A unified and multi-level evaluation framework is rarely applied. Even when multiple metrics are used, most operate at the level of individual utterances or predefined personality traits, while paying limited attention to the linguistic form through which character identity is expressed. As Tseng et al. [2024] note in a recent survey, despite the growing number of evaluation approaches, "there still lacks a unifying understanding of how to quantify personality in LLMs". In addition, existing evaluations seldom examine which metrics most reliably capture character consistency across extended interactions.

To address this limitation, the present study develops a set of metrics grounded in the linguistic hierarchy and validates them through classification performance and confidence calibration. The approach proposed in this thesis differs from existing evaluation frameworks in an important way. Rather than focusing primarily on high-level behavioural outcomes or abstract personality traits, it examines how character consistency is realised through linguistic structure during dialogue. Specifically, the analysis operates across three layers of language: lexical, syntactic, and discourse features.

Most existing studies evaluate role-playing agents through macro-level indicators, such as personality traits, value alignment, or task success. While these approaches provide useful insights, they often overlook how character identity is realised through language during ongoing interaction [Shao et al., 2023, Wang et al., 2024c, Wu et al., 2024]. The method proposed in this thesis instead investigates the micro-level linguistic structure of agent dialogue, allowing subtle shifts in speaking style to be detected over the course of extended collaborative interactions.

At the lexical level, the analysis can reveal changes in the use of character-specific vocabulary or recurring expressions. Such patterns may indicate whether the agent continues to employ a distinctive character voice [Zhou et al., 2023, Wang et al., 2024a].

At the syntactic level, structural features of sentences can signal shifts in communicative style. For instance, a character originally designed to speak in analytical or elaborate sentences may gradually revert to shorter and more generic responses typical of default assistant behaviour [Park et al., 2023, Wang et al., 2024a].

At the discourse level, the analysis examines how the agent maintains conversational coherence and fulfils its narrative role within collaborative dialogue. This includes how the agent manages topic development, responds to other participants, and sustains the interactional logic of the scenario [Ji et al., 2025].

Although recent work has begun to discuss phenomena such as identity drift or persona drift in interactive agents, these studies often rely on isolated psychological measurements or discrete evaluation snapshots. They rarely track linguistic behaviour continuously throughout the dialogue itself. By contrast, the approach adopted in this thesis provides an intrinsic, corpus-based evaluation framework. Through systematic measurement of lexical, syntactic, and discourse-level features, it offers a new quantitative perspective on how character consistency evolves during long-term interaction [Choi et al., 2024].

2.5 LLM-based Multi-agent Collaboration

2.5.1 The Evolution Of LLM-based Multi-agent Systems

The transition from single-agent systems to Multi-Agent Systems (MAS) marks a pivotal shift in LLM applications, moving from individual task execution to the emulation of collective intelligence [Guo et al., 2024, Xi et al., 2025]. By assigning specialized profiles and enabling inter-agent communication, MAS can effectively mirror human group dynamics in complex problem-solving scenarios, such as scientific debate, software development, and detective reasoning [Guo et al., 2024].

This study choses the detective case-solving scenario because it represents a pinnacle of complex social reasoning, requiring agents to not only process factual evidence but also to perform Theory of Mind (ToM) inferences regarding the motives and beliefs of other characters [Zhu et al., 2024]. Recent frameworks like WellPlay [Zhu et al., 2024] and Jubensha [Wu et al., 2024] have established benchmarks for measuring the task accuracy and reasoning efficiency of agents in such settings.

However, a significant gap remains in the literature. While current research heavily prioritizes optimization of cooperation strategies and final task outcomes (e.g., win rates), it often overlooks the linguistic integrity of the personas under the pressure of collaboration [Chen et al., 2024b, Wu et al., 2024, Zhu et al., 2024]. The very nature of multi-round interaction introduces risks of Identity Drift and Persona Inconstancy, where agents may succumb to peer pressure or inadvertently adopt the linguistic patterns of their counterparts [Choi et al., 2024, Baltaji et al., 2024]. Unlike existing works that rely on macro-level performance metrics, this thesis shifts the focus to the micro-linguistic level. By analysing dialogues through lexical, syntactic, and discourse metrics, this study aims to quantify the subtle "linguistic imprint" left by collaboration and detect whether the intense interaction required for problem-solving inevitably leads to character inconsistency.

2.5.2 Identity Contamination In Multi-agent Collaboration

Finally, the special dynamics of multi-agent systems is also worth elaborated. When multiple LLM agents interacts, linguistic phenomena such as alignment and convergence can occur. Recent surveys on multi-agent systems (MAS) highlight that while specialized profiles and inter-agent communication facilitate collective intelligence [Guo et al., 2024], they also introduce complex social dynamics [Zhu et al., 2024]. This includes the risk of information redundancy and "over-reporting," where agents may prioritize group alignment over individual task precision [Guo et al., 2026]. Just as humans in conversation tend to adapt to each other's style (accommodation), groups of AI agents may inadvertently begin to share vocabulary or syntax. Recent work by Parfenova et al. [2025] demonstrates this: in simulated collaborative annotation tasks, LLM groups were found to lexically and semantically converge over multi-round discussions.

This linguistic convergence poses a severe risk of identity drift and identity contamination. Choi et al. [2024] define identity drift as the phenomenon where interaction patterns or styles change over time, noting that larger models may paradoxically struggle more with identity stability due to their tendency to generate fictitious self-details. Furthermore, Baltaji et al. [2024] identify "Persona Inconstancy" as a critical failure mode in cross-national collaborations, categorising it into conformity, confabulation, and the most disruptive form—impersonation, where an agent abandons its assigned identity to adopt another mentioned in the chat context. Such lapses often stem from the lack of clear separation between role descriptions and the growing linguistic context of the dialogue.

Moreover, the singular persona convergence observed in models fine-tuned via RLHF often leads agents to default to a generic, polite assistant style, eroding the specific values and traits required for complex role-playing [Park et al., 2023, Moon et al., 2024]. Such contamination reduces the diversity and specificity of personas, which undermines the goal of simulating heterogeneous characters. This study argues that traditional macro-level evaluations (like task accuracy) are insufficient to capture these micro-linguistic shifts. By employing lexical, syntactic, and discourse-level metrics, this study provides a continuous monitoring tool for character (in)consistency, helping to ensure that multi-agent dialogues preserve the individuality of each character even under intense collaboration.

3 Methods

This study adopts a simulation-based experimental design to examine character consistency and collaborative reasoning in large language model (LLM)–driven agents. A controlled multi-agent dialogue environment was developed in which three agents, each role-playing a canonical fictional detective, collaboratively solve a murder case through turn-based natural language interaction. The simulation prioritises reproducibility, interpretability, and detailed logging to enable systematic analysis.

The fictional detectives included in the simulation are **Sherlock Holmes** from Arthur Conan Doyle and **Miss Marple** and **Hercule Poirot** from Agatha Christie. These characters were selected for their distinctive personalities, investigative approaches, and differing genders. Moreover, their extensive and well-documented literary corpora provide rich linguistic material, making them particularly suitable for systematic analysis.

3.1 Prompt Engineering

This study employs prompt engineering as the central mechanism for constructing non-parametric Role-Playing Language Agents (RPLAs). Rather than enhancing task performance through external knowledge augmentation, the aim is to investigate how character-specific descriptions which encompassing both static attributes (e.g., identities, viewpoints) and dynamic behaviours (e.g., linguistic features, interaction patterns) shape language generation and collaborative reasoning [Zhou et al., 2023, Chen et al., 2024b].

Advanced architectures such as Retrieval-Augmented Generation (RAG) or specialised long-term memory modules, were intentionally excluded from this study. Their inclusion would introduce significant confounding factors by merging external information streams with the agent's internal parametric knowledge, making it impossible to isolate the influence of prompt-based conditioning on agent behaviour [Liu et al., 2023, Guo et al., 2024, Xi et al., 2025]. By adopting a prompt-only setup, this study aims to ensure that any observed instances of character hallucination (the intrusion of out-of-character knowledge) [Shao et al., 2023] or persona inconstancy (such as conformity or impersonation) [Baltaji et al., 2024] can be directly attributed to the model's processing of the provided instructions within the evolving dialogue context [Chen et al., 2024b].

Furthermore, this approach facilitates a rigorous analysis of linguistic fidelity [Wang et al., 2024a]. By limiting the system to prompt-based conditioning, this study can more effectively measure how lexical consistency and dialogic fidelity (e.g., syntactic tone) fluctuate under the pressure of multi-agent collaboration

[Wang et al., 2024a, Ji et al., 2025]. This design aims to preserve high internal validity, shifting the focus from system optimisation to the micro-level quantification of identity drift—a phenomenon where interaction patterns and styles change over time due to the influence of the conversational history Choi et al. [2024]. Consequently, prompt engineering offers a transparent and traceable framework for reproducibility, allowing for detailed post-hoc inspection of how linguistic constraints dictate the boundaries of LLM-simulated personas [de Lima et al., 2025].

3.2 Model Selection And Inference Configuration

To ensure that any observed variations in Character Consistency arise purely from prompt-level conditioning rather than model-specific biases, all agents in this study are powered by the same underlying large language model. This study utilises Meta's **Llama-3.2 3B Instruct** (`meta/llama-3.2-3b-instruct`), a lightweight yet capable instruction-tuned transformer model comprising approximately three billion parameters. The model is accessed as a non-parametric backbone via the NVIDIA API endpoint through the `langchain nvidia ai endpoints` interface, ensuring a high level of control over the reasoning core without the confounding effects of model fine-tuning or parametric updates [Shao et al., 2023].

To preserve experimental rigour and minimise stochastic noise that could obscure the comparative analysis of linguistic behaviour, all inference parameters are held constant across all agents and cases. Following the experimental setups in recent character drift studies, this study sets nucleus sampling (top-p) to 0.9 and maximum token length to 512 to ensure sufficient response depth for collaborative reasoning [Choi et al., 2024]. While some studies prefer deterministic outputs (temperature = 0.0) for extraction tasks [de Lima et al., 2025], this study sets the temperature to 0.7 to "mirror real-world usage" and elicit a more natural and diverse range of conversational patterns necessary for detecting subtle linguistic shifts [Choi et al., 2024]. Fixed request timeouts (60 seconds) and single-response generation (n=1) are maintained to ensure system stability across multi-round detective simulations.

Each agent interacts with the model through a unified conversational interface where the model operates in a stateless manner at each invocation [Park et al., 2023, Guo et al., 2026] . Unlike systems with internal long-term memory modules that might introduce hidden state carry-over, conversational continuity is maintained externally through explicitly controlled prompt construction. By managing dialogue history in the context window and strictly partitioning role descriptions from chat history, this design effectively isolates the linguistic effects of prompt conditioning [de Lima et al., 2025]. This isolation is critical for our multi-level analysis, as it ensures that any erosion of character integrity—whether at the lexical, syntactic, or discourse level—can be directly traced to the dynamics of the multi-agent collaboration itself [Choi et al., 2024, Guo et al., 2026].

3.3 Experimental Scenarios And Case Selection

This study adopts three murder mystery cases originally developed for the Jubensha benchmark [Wu et al., 2024], a Chinese detective role-playing game framework designed for multi-agent interaction. Unlike traditional single-agent reasoning tasks, these cases provide a complex, plot-driven environment that requires agents to perform Theory of Mind (ToM) inferences and negotiate over contradictory evidence [Zhu et al., 2024, Wu et al., 2024]. While the original framework focused on task completion, this study restructures these cases into a controlled experimental setting to isolate the linguistic markers of Character Consistency.

The decision to use pre-existing, human-authored Jubensha cases rather than canonical literary mysteries (e.g., from the works of Arthur Conan Doyle or Agatha Christie) is a deliberate choice for experimental rigour. Recent studies indicate that LLMs possess "varying degrees of familiarity with detective narratives" through their pre-training data [de Lima et al., 2025]. Using famous stories like The Adventure of the Speckled Band would introduce a high risk of Data Leakage or Character Hallucination, where agents might derive a solution based on memorized plots rather than collaborative reasoning [Chen et al., 2023b, 2024b]. By using "neutral" and culturally diverse cases from the Jubensha dataset, this study ensures that the agents' responses are driven by the provided non-parametric context rather than parametric memory, thereby preserving the internal validity of our analysis on character-specific linguistic drift.

To support systematic control over dialogue dynamics, each case was converted into a structured YAML template. This format decomposes the narrative into distinct modules: a global setting, victim profiles, suspect attributes (including motives and secrets), crime scene data, forensic evidence, and a chronological timeline [Wu et al., 2024, Zhu et al., 2024]. This modularity allows us to enforce Information Asymmetry across the three agents, mirroring the Social Reasoning challenges found in human group dynamics [Guo et al., 2026]. During the simulation, specific information subsets are assigned based on the detectives' established archetypes [de Lima et al., 2025]:

- Sherlock Holmes is provided with forensic and crime scene data, reflecting his evidence-driven, analytical method.
- Hercule Poirot is assigned the timeline and procedural records, reflecting his methodical reasoning and mental synthesis.
- Miss Marple receives suspect profiles and interpersonal relations, reflecting her reliance on social observation and human behaviour parallels.

The three selected cases vary in suspect density (four, four, and six suspects, respectively) to introduce incremental levels of Reasoning Complexity and social entropy [Baltaji et al., 2024, Guo et al., 2026] . By mapping the original scripts into this template-based design, this study creates a reproducible environment where the pressure to solve the crime and the subsequent need for Information Sharing act as a linguistic "stress test" for Character Consistency [Park et al., 2023, Guo et al., 2026]. This setup allows precisely track whether the demand for group consensus leads to a convergence of styles or a breakdown of the unique "investigative fingerprints" assigned to each detective.

3.4 Agent Construction: Character Simulation

To investigate Character Consistency under collaboration, three distinct detective agents were constructed by prompting. Each agent is implemented as an instance of a shared `BaseAgent` class, which encapsulates the language model interface and a lightweight text-based memory module. This architectural design ensures structural homogeneity across the system, guaranteeing that any observed variation in character arises solely from prompt engineering and differential information exposure rather than model architecture or parameter disparity.

Each agent's prompt is dynamically regenerated for every model call and is composed of five modular components (See Appendix for the full prompt):

1. **Background and Identity**: This component establishes the shared "closed-room" scenario and explicitly assigns the character's identity (e.g., "You are Sherlock Holmes"). Following the design principles of Character-GLM [Zhou et al., 2023], these "Static Attributes" define the core knowledge and investigative methods inherent to the character's archetype.

2. **Collaboration Rules**: This module defines global behavioural constraints, requiring agents to reason strictly from stated evidence and dialogue history, avoid fabricating new clues, and maintain transparency by sharing all received information. These rules promote collective intelligence through disciplined deductive interaction rather than isolated speculation.

3. **Role-play Guidelines** (Dynamic Behaviours): This section describes each detective's persona from a linguistic perspective, ensuring Linguistic Fidelity [Wang et al., 2024a]. Inspired by de Lima et al. [2025], there are five linguistic aspects covered: (1) vocabulary, (2) lexical features, (3) syntactic features, (4) discourse patterns, and (5) investigative methods. For instance, Sherlock Holmes is characterised by scientific vocabulary, logically nested sentences, and a detached analytical tone. The steps of completing this is:

Step 1 **Generation**: Following the methodology in de Lima et al. [2025], a GPT-based API (`GPT-4`) was employed to generate the initial character profiles and final role-play prompts. During this automated generation phase, the temperature was fixed at 0.0 to ensure deterministic and consistent persona descriptions. For each description in the role-play prompts, an example is manually extracted from the original novels.

Step 2 **Validation**: To assess the validity of these representations, a *Validation Via Reverse Identification* is conducted following de Lima et al. [2025], requiring an independent LLM (Llama-3.2-3B-Instruct) to correctly infer the detective's identity based solely on the generated linguistic descriptions.

4. **Protective Guidelines**: To preserve character integrity over extended dialogue, we implemented strict behavioral prohibitions designed to prevent stylistic drift. For example, Holmes is instructed to avoid emotional expression or social small-talk. These constraints serve as guardrails against Character Hallucination, which is the tendency of LLMs to default to a generic "AI assistant" style that violates their assigned persona [Shao et al., 2023, Chen et al., 2024b].

5. **Information Allocation**: Based on the detectives' established investigative archetypes, each agent is assigned a unique subset of case information from the YAML template as stated above. This deliberate *Information Asymmetry* enforces interdependence, compelling agents to exchange unique clues to reach a coherent collective conclusion [Zhu et al., 2024].

3.5 Simulation Procedure: Collaborative Reasoning

The simulation is implemented as a turn-based, multi-agent dialogue where the three detective agents engage in complex social reasoning to solve a murder case. Each experimental run is initialised with an empty shared dialogue memory. The session acts as a "stress test" for character consistency, as agents must balance the preservation of their unique persona with the demand for group consensus [Chen et al., 2024a].

The procedure follows a rigorous cycle of interaction and belief monitoring:

- **Controlled Interaction Flow**: At the start of each round, the speaking order of the three detectives is randomly permuted. This serves to mitigate positional bias and the Initiator Dominance effect, where the first speaker's opinion might disproportionately influence the group's trajectory.

- **Linguistic Protocol Enforcement**: This study implements a strict one-speaker-per-turn protocol. Raw model outputs are post-processed to remove redundant speaker prefixes and truncate any content where the model attempts to self-impersonate or generate dialogue for other characters, thereby maintaining clear character boundaries [Chen et al., 2024b].

- **Belief State Alignment Tracking**: To quantitatively monitor the evolution of deductive logic, each agent is required to end every utterance with an explicit hypothesis in a fixed pattern: "I believe the murderer is [Suspect]". This constraint enables automatic Belief State Extraction without semantic parsing, allowing us to track exactly when and how character consistency might erode in favour of Conformity [Baltaji et al., 2024].

- **Termination and Logging**: The simulation terminates when Belief State Alignment is achieved (all three agents independently nominate the same suspect) or when the limit of ten rounds is reached.

Throughout the process, the system maintains comprehensive logging for post-hoc analysis. Surface-level dialogue is recorded in CSV format (including turn index, speaker, and extracted hypothesis), while the raw, dynamically evolving prompts for each turn are stored. This dual-logging design ensures full observability and reproducibility, providing a structured basis for our downstream analysis of linguistic drift across lexical, syntactic, and discourse levels.

4 Evaluation Scheme

The evaluation scheme is designed to rigorously assess the trade-off between task utility (solving the mystery) and persona fidelity (remaining in character) within a multi-agent detective simulation. Given that the collaborative pressure to reach a consensus often triggers identity contamination or character drift, this study treats character consistency as a multidimensional construct and move beyond macro-level success rates to implement a fine-grained analysis of Out-of-Character (OOC) behaviours [Zhou et al., 2023], quantified through specific linguistic markers across three hierarchical levels.

To ensure robustness, the evaluation framework integrates task-level outcome metrics, model-based classification as baseline, and a suite of quantitative linguistic measures to track the evolution of character integrity turn-by-turn.

4.1 Evaluation Overview

Each simulation run is evaluated across two primary axes, reflecting the agent's dual role as a problem-solver and a role-playing entity.

1. **Task Performance (Utility)**: This study measures the collective intelligence of the detective trio by their ability to accurately identify the murderer within the allotted rounds, similar to the win rate in Zhu et al. [2024]. This axis assesses whether the specific persona prompts or the dynamics of collaboration facilitate or hinder the logical reasoning required for the mysterious cases.

2. **Character Consistency (Linguistic Fidelity)**: This study quantifies the stability of each agent's "investigative fingerprint" throughout the dialogue. Rather than relying on a single aggregate score, character consistency is operationalised through multiple measures spanning lexical, syntactic, and discourse levels.

In addition, to establish a robust baseline for evaluating character consistency, a fine-tuned BERT-base-cased text classification model is employed. This model functions as an automated "Style Discriminator" to determine the probability that a given utterance belongs to its assigned detective persona [Wang et al., 2024b].

All evaluations are conducted turn-by-turn, enabling both static comparisons against reference texts and dynamic analyses of consistency development over the course of a dialogue.

4.2 Task Performance

To evaluate the utility of the multi-agent system, task performance is quantified through murderer identification accuracy across three distinct experimental conditions. This ablation-style design allows to isolate the individual contributions of character-specific constraints and collaborative instructions on the agents' deductive reasoning:

- Condition 1: **Zero prompt (Baseline)**: Agents are tasked with solving the case without specific character profiles or collaboration guidelines, serving as a measure of the LLM's inherent reasoning capacity.
- Condition 2: **Character prompts only**: Agents are provided with the detectives' profiles but no explicit instructions on how to collaborate. This condition tests whether the adoption of a customised characterisation independently enhances or hinders task performance.
- Condition 3: **Collaboration prompt only**: Agents receive rules for group interaction (e.g., sharing clues, reaching consensus) but are not assigned character prompts. This focuses on the effect of collective intelligence and the potential risk of conformity in the absence of character-specific boundaries.

4.3 Classification Baseline

A supervised text classification model is employed as a reference-based measure of character consistency. The classifier is trained on dialogue excerpts extracted from the original literary works featuring Sherlock Holmes, Hercule Poirot, and Miss Marple, all of which are in the public domain. The objective is to construct a probabilistic model of character voice that can serve as an external stylistic benchmark against which simulated utterances are evaluated.

For each simulation turn, the classifier outputs a probability distribution over character labels. The probability assigned to the intended character is interpreted as a continuous measure of character consistency. This enables analysis of both absolute consistency levels and temporal trends across dialogue turns.

4.3.1 Data Construction

Character utterances were extracted using an LLM-assisted pipeline to improve speaker attribution reliability. The initial corpus comprised six recurring characters: in additional to the detective trio, three leading characters are also selected: Dr. John Watson from Sherlock Holmes's stories, Inspector James Japp and Captain Arthur JM Hastings from Hercule Poirot's stories. Table1 summarises corpus size and data quality indicators prior to cleaning.

Character	Utterances	Tokens	Avg Tokens per Utterance	Empty / Noise	< 3 Tokens	Duplicates
Holmes	5,509	129,893	23.58	10.35	12.36	14.30
Marple	6,551	148,602	22.68	11.16	13.05	18.29
Poirot	6,758	145,881	21.59	8.27	9.32	15.73
Watson	5,186	118,378	22.83	10.16	12.36	16.14
Hastings	2,979	59,027	19.81	11.58	13.29	18.63
Japp	1,584	32,740	20.67	18.56	20.14	24.05

Table 1: Initial Corpus Statistics and Data Quality Diagnostics (% of Utterances)

Across the corpus, average utterance length ranged between approximately 20 and 24 tokens. Fewer than 2% of utterances exceeded 128 tokens, indicating minimal truncation risk under the model's input constraints.

A diagnostic analysis revealed that Japp exhibited the highest combined rate of empty utterances, extremely short utterances, and duplication, alongside the smallest corpus size. Given this elevated noise profile and limited data volume, Japp was excluded from subsequent training to preserve data quality and maintain class balance.

4.3.2 Cleaning And Balancing

For the remaining characters, a cleaning procedure was applied. Empty utterances, placeholder strings, utterances shorter than three tokens, and exact duplicates within each character were removed.

After cleaning, datasets were balanced by total token count rather than by utterance count. This decision was made to reduce systematic bias arising from variation in utterance length across characters. Balancing by token count ensures that each class contributes comparable lexical volume during training.

4.3.3 Model Architectures And Training Procedure

The classifier is implemented using with a task-specific classification head operating on the pooled CLS representation. The model used is `BERT-base-cased`. This BERT-base classifier was trained under two configurations.

The first configuration was a three-class model distinguishing Holmes, Poirot, and Marple. Each character contributed approximately 120,000 to 130,000 tokens after balancing.

The second configuration was a four-class model including an additional heterogeneous "Other" category composed of Watson and Hastings. In this setting, each class contained approximately 50,000 tokens.

For both configurations, utterances were fully shuffled prior to dataset splitting. An 80/10/10 train, validation, and test split was applied. Models were trained with a batch size of 8, a learning rate of 2e-5, and weight decay of 0.01.

The three-class model was trained for both one and three epochs to examine convergence behaviour. The four-class model was trained for three epochs.

4.3.4 Baseline Performance Comparisons

The classification results are summarised in Table2 below.

Model	Epochs	Train Accuracy	Test Accuracy	Test Macro-F1
3-class	1	0.82	0.74	0.74
3-class	3	0.89	0.76	0.76
4-class	3	0.81	0.55	0.54

Table 2: Comparisons of Baseline Performance

The three-class model achieved stable performance after a single epoch. Additional training produced only marginal improvements, suggesting rapid convergence and robust baseline separability among the three target characters.

In contrast, the four-class model exhibited substantially lower test accuracy and macro-F1. The heterogeneous "Other" category likely introduced internal variability that reduced class separability. This configuration therefore provided weaker discriminative performance.

On the basis of these results, the three-class model trained for three epochs was selected as the primary classifier for subsequent character-consistency analyses. This configuration provides stable performance and interpretable probabilistic outputs while maintaining clear class boundaries among the three detective characters.

4.4 Lexical Metrics

Lexical character consistency is evaluated using distributional **vector-based similarity** measures. For each generated utterance, lexical similarity to the corresponding character's gold-standard corpus (Holmes, Poirot, Marple) is computed using cosine similarity between TF-IDF representations. This comparison captures whether an agent's vocabulary usage remains aligned with the characteristic lexical patterns of the target character or shifts toward generic or cross-character language. In this way, the metric provides an estimate of how strongly the generated dialogue reflects character-specific lexical style.

However, lexical similarity alone may be influenced by shared conversational topics. In collaborative reasoning tasks, multiple agents often discuss the same evidence or suspects, which can artificially increase similarity across speakers. To control for this effect, the analysis introduces an additional **intra-agent lexical similarity** framework that explicitly contrasts within-character continuity with cross-character similarity at the same dialogue turn.

The corrected metric consists of three components. First, **intra-agent similarity** measures the cosine similarity between consecutive utterances produced by the same character, capturing lexical continuity within that agent's dialogue.

Second, **cross-agent similarity** measures the average cosine similarity between the current utterance of a character and the utterances produced by the other agents at the same turn index. This provides an estimate of lexical similarity that may arise from shared discussion topics rather than character identity.

Based on these two quantities, a **character distance score** is computed:

$$\text{character\_distance} = \text{intra\_agent\_similarity} - \text{cross\_agent\_similarity} \quad (1)$$

This difference quantifies whether lexical similarity is more strongly associated with the character's own previous speech or with the concurrent dialogue of other agents. Positive values indicate that lexical continuity is primarily driven by character-specific language use, whereas values close to zero suggest topic-driven similarity across agents. Negative values indicate that the current utterance resembles the language of other agents more strongly than the character's own previous speech, which may signal potential out-of-character behaviour.

All similarity computations are performed using TF-IDF representations constructed over the full set of dialogue utterances in the simulation logs. This choice maintains consistency with the reference-based lexical similarity metric and provides an interpretable representation of vocabulary distribution.

4.5 Syntactic Metrics

Syntactic character consistency is evaluated through dependency parsing, focusing on **dependency tree depth** as an indicator of structural complexity. This measure captures the degree of hierarchical organisation in a sentence and has been widely used as a proxy for syntactic complexity. In this study, dependency depth is used to examine whether LLM-generated dialogue maintains the characteristic syntactic style associated with each fictional detective.

A character-specific reference distribution of syntactic depth is first constructed from the literary corpora corresponding to each detective. To ensure that the reference style reflects meaningful syntactic structure rather than short conversational fragments, baseline utterances are filtered prior to sampling. By default, utterances shorter than a minimum token length are removed before random sampling, although an alternative filtering strategy based on minimum dependency depth may also be applied. For each character, a fixed number of filtered utterances are sampled and parsed using the spaCy dependency parser (`en_core_web_sm`). Sentence-level dependency depth is computed recursively from root nodes, and utterance-level syntactic complexity is defined as the mean depth across all sentences in the utterance. These values form the character-specific reference distribution.

Using this reference distribution, two types of standardised deviation scores are computed for each utterance in the simulation dialogue. The first metric is the **reference-based z-score** (z_{ref}), which measures the deviation of an utterance's syntactic depth from the canonical reference style of the corresponding character. Formally,

$$z_{\text{ref}} = \frac{\text{depth} - \mu_{\text{ref}}}{\sigma_{\text{ref}}} \tag{2}$$

where μ_{ref} and σ_{ref} denote the mean and standard deviation of syntactic depth in the character's baseline corpus. This metric answers the question: *to what extent does a generated utterance deviate from the original syntactic style of the character?*

In addition to reference-based comparison, an internal standardisation is computed within the simulation corpus. For each character, the mean and standard deviation of syntactic depth across all generated utterances are used to calculate a **simulation-based z-score** (z_{sim}). This second metric indicates whether a particular utterance is syntactically extreme relative to other utterances produced by the same agent during the simulation.

Together, these two measures allow the analysis to distinguish between two types of syntactic variation: deviation from the canonical literary style (z_{ref}) and internal variability within the simulated dialogue (z_{sim}). Formally,

$$z_{sim} = \frac{depth - \mu_{sim}}{\sigma_{sim}} \tag{3}$$

where μ_{sim} and σ_{sim} denote the mean and standard deviation of syntactic depth computed over all utterances produced by the same character in the simulation corpus.

Finally, to examine whether syntactic behaviour changes systematically during extended interaction, **syntactic inconsistency** is analysed using regression models over dialogue turns. For each character, utterances from all simulation runs are pooled and linear regressions are fitted with turn index as the independent variable and syntactic deviation as the dependent variable. The primary analysis uses z_{ref} as the response variable in order to detect gradual divergence from the reference syntactic style. As a robustness check, a parallel regression using z_{sim} is also conducted to capture internal syntactic drift within the simulation corpus. Positive or negative slopes indicate systematic increases or decreases in syntactic complexity relative to the baseline style over the course of the dialogue.

All syntactic measurements are computed using the same dependency parsing pipeline for both reference and simulation corpora to ensure methodological consistency. The resulting per-utterance metrics and aggregated statistics form the basis for subsequent analysis of character-level syntactic consistency.

4.6 Discourse Metrics

At the discourse level, this study analyses **sentiment trajectories** as an auxiliary indicator of character consistency. Sentiment is measured using VADER, which produces four-dimensional polarity vectors (positive, negative, neutral, compound) for each utterance.

First, reference sentiment profiles are constructed from the original literary corpora by averaging VADER sentiment vectors for each character. These reference vectors represent the baseline emotional tendencies associated with each persona.

For simulation data, sentiment is computed for each utterance and aggregated at the turn level for each character. Then **sentiment distance** is measured as the Euclidean distance between the turn-level sentiment vector and the corresponding character's reference vector. Larger distances indicate stronger deviations from the canonical sentiment profile.

How sentiment distance evolves over dialogue turns is then analysed to examine potential discourse-level drift. For each character within a simulation run, linear regression is applied with turn index as the independent variable and sentiment distance as the dependent variable. The resulting slope captures the direction and magnitude of sentiment drift across the interaction.

Summary statistics are further aggregated across characters and scenarios to compare average inconsistency patterns. While sentiment in many studies is not treated as a primary marker of persona identity, these trajectories provide an additional discourse-level signal for detecting potential inconsistencies in character behaviour over time.

4.7 Validation Across Metrics

To examine whether the proposed linguistic metrics capture character consistency in a meaningful way, they are validated against the classifier-based consistency scores. The classifier predicts the probability that an utterance belongs to the correct character. This probability is used as a reference signal for character consistency.

All evaluation metrics are first aligned at the turn level. For each case, turn, and character, the outputs of the lexical, syntactic, and discourse metrics are merged with the classifier predictions. When multiple observations are available within the same turn, values are averaged to obtain a single turn-level representation.

For each metric, correlations with the classifier probability using both Pearson and Spearman coefficients are computed. Pearson correlation measures linear association, while Spearman correlation captures monotonic relationships. Strong correlations suggest that a metric tracks variations in character consistency in a way that is broadly aligned with the classifier signal.

Calibration between linguistic metrics and classifier confidence is also examined. Metric values are first normalised to the range $[0, 1]$ using min–max scaling. Expected Calibration Error (ECE) is then computed by comparing the scaled metric values with the classifier probabilities across confidence bins. Lower ECE indicates better alignment between metric scores and classifier confidence.

Finally, the metrics are compared using a composite *faithfulness score*. This score combines the absolute Pearson correlation, the absolute Spearman correlation, and the calibration score derived from ECE. All components are normalised to the range $[0, 1]$ before averaging. Metrics with higher faithfulness scores are interpreted as more reliable indicators of character consistency.

4.8 Clustering-based Contamination Analysis

Character contamination across agents is further examined using clustering analyses. Utterances from both the reference corpus and the simulation dialogues are represented as vectors and grouped into three clusters corresponding to the three detectives. Clustering quality is evaluated using standard separability metrics, including silhouette coefficients, cluster purity, and adjusted Rand index (ARI). Comparisons between reference and simulation data provide an aggregate indication of whether collaborative dialogue reduces character distinctiveness.

Two types of vector representations are used. First, embedding vectors are extracted from the encoder of the fine-tuned three-class character classifier, using the CLS hidden states as utterance representations. Clustering results are evaluated against the true character labels ($\text{ARI}_{\text{character}}$) and, for simulation data, also against case identifiers (ARI_{case}) to assess whether clustering reflects narrative context rather than character identity.

Second, a metric-based representation is constructed by concatenating turn-level linguistic evaluation metrics into feature vectors. These include lexical similarity, intra-agent character distance, syntactic deviation (z_{ref}), syntactic depth, and sentiment distance, with optional inclusion of additional internal metrics such as z_{sim} and intra-agent similarity. This representation provides an alternative perspective on character separability based on the behavioural signals captured by the proposed linguistic measures.

Clustering performance on simulation data is then compared with results obtained from the reference corpus. If collaborative simulation reduces cluster separability relative to the reference texts, this is interpreted as evidence of cross-character stylistic leakage and increased out-of-character behaviour.

5 Results

5.1 Data Overview

Before presenting the experimental results, an overview of the reference corpus and simulation data used in the analyses is presented.

The **reference corpus** consists of 13,733 utterances extracted from canonical texts. The distribution across characters is relatively balanced: 4,907 utterances for Poirot, 4,445 for Marple, and 4,381 for Holmes. This corpus forms the basis for training and validating the character classifier and for constructing linguistic reference profiles.

The **simulation dataset** comprises 387 utterances generated across 30 multi-agent runs (three cases $\times$ ten runs per case). At the case level, utterances were distributed as follows: 147 for Case 1, 120 for Case 2, and 120 for Case 3.

This relatively balanced simulation design supports fair comparison across characters and cases in subsequent linguistic and clustering analyses.

5.2 Task Performance

To evaluate the influence of character-specific prompting and collaborative dynamics on the deductive capabilities of LLM agents, the murderer identification accuracy is calculated across repeated simulation runs (N=50 for each case/-condition). The results summarised in Table 3 reveal a distinct **performance gradient** where task utility is progressively challenged by the introduction of persona constraints and group interaction.

Condition	Case 1	Case 2	Case 3
C1: Zero-shot (Baseline)	97% (47/50)	100%	100%
C2: Persona Prompt (No Collaboration)	92% (46/50)	36% (18/50)	100%
C3: Collaboration Prompt (No Persona)	20% (10/50)	50%	40%

Table 3: Murderer Identification Accuracy across Three Experimental Conditions

5.2.1 Baseline Reasoning Capacity (C1)

In the zero-shot baseline condition, the agents exhibited exceptional deductive performance, achieving near-perfect accuracy (97% to 100%) across all three cases. These results confirm that the underlying LLM possesses the necessary parametric knowledge and reasoning depth to solve these cases when acting as a stand-alone generic reasoner [Wu et al., 2024]. This high baseline serves as the upper-bound performance reference and suggests that any subsequent decline in accuracy is not due to an inherent lack of reasoning ability but rather the cognitive load introduced by the character-specific prompting or collaborative dynamics.

5.2.2 The Impact Of Character Prompts (C2)

The introduction of character prompts in Condition 2 (without collaboration) led to significant variability in performance. While Case 3 remained unaffected (100%), Case 1 saw a slight decline to 92%, and Case 2 suffered a substantial performance collapse to 36%.

This discrepancy suggests that **character consistency exists in a trade-off with task utility**. In Case 2, the specific role-playing required by the character prompt may have acted as a distraction. As noted in the literature, enforcing a specific character identity can hinder a model's ability to utilize its full reasoning capacity, especially when the character's "voice" or "behavioural constraints" conflict with the most direct logical path to the solution [Shao et al., 2023, Samuel et al., 2024].

5.2.3 The Impact Of Collaborative Dynamics (C3)

The most striking finding is the sharp reduction in accuracy across all cases when multi-agent collaboration was introduced. Despite the agents having access to partial case clues which is designed to encourage collective intelligence, accuracy plummeted to 20% for Case 1, 50% for Case 2, and 40% for Case 3. Comparing the C1 and C2, the collaborative setting yielded the lowest performance levels.

As noted in previous work, this collaboration paradox can be interpreted through several theoretical lenses.

First, agents demonstrate extreme sensitivity to perceived peer pressure, resulting in strong conformity during discussions. This often leads them to sacrifice individual logical rigour or their pre-assigned investigative stance in favour of achieving alignment with the group [Baltaji et al., 2024].

Second, the detective scenario involves a complex social reasoning bottleneck, requiring agents to possess high-order Theory of Mind (ToM) inference capabilities to track the beliefs and hidden goals of others. However, existing collaboration mechanisms often trigger information redundancy and noisy dialogue streams, which can obscure key evidence and hinder the deductive process [Wu et al., 2024, Zhu et al., 2024, Guo et al., 2026].

5.2.4 Case-specific Sensitivity

Overall, these results demonstrate a clear performance gradient across conditions, with highest accuracy observed in the zero-shot baseline, intermediate and case-dependent performance under persona prompting, and lowest accuracy in the collaborative condition. This suggests that the impact of character persona and collaboration is not uniform but highly sensitive to case-specific characteristics.

5.3 Classifier-based Model Baseline

A BERT-based text classification model is trained as a model-based baseline to evaluate character consistency, as describe in the *Methods* section. To evaluate whether character identity becomes less recognisable as dialogue progresses, classifier confidence as a function of turn index is modelled. Specifically, the probability assigned by the trained classifier to the correct character label (`prob_correct`) is extracted for each utterance. Linear regressions were then conducted with turn index as predictor. Negative slopes indicate declining attribution confidence over time and are interpreted as evidence of behavioural drift. In addition, the mean squared error of predicted probabilities - Brier scores (`brier_score`) - were examined descriptively to assess overall calibration stability.

Regression analyses were conducted separately for each character within each case, pooling utterances across runs in order to estimate character-level temporal trends. Even where slopes did not reach statistical significance, their direction and magnitude are reported to provide a fuller picture of temporal tendencies. Although statistically significant effects are noted, explained variance remains modest ($R^2 \leq 0.15$ in all cases), indicating that turn index accounts for only a limited proportion of variance in attribution confidence.

The results are presented collectively for clearer comparisons and overview. Table 4 and Table 5 below show the regression results for predicted probability of correct class and brier score over turns.

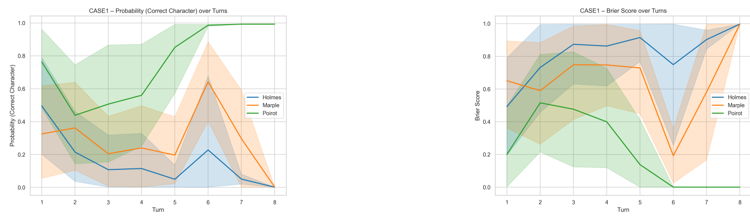
Case	**Character**	β **(prob_correct)**	**95% CI**	R^2
case1	Holmes	-0.063*	[-0.118, -0.008]	0.102
case1	Marple	-0.002	[-0.064, 0.060]	0.000
case1	Poirot	0.057	[-0.009, 0.124]	0.060
case2	Holmes	-0.009	[-0.080, 0.061]	0.002
case2	Marple	-0.040	[-0.101, 0.021]	0.045
case2	Poirot	-0.035	[-0.104, 0.034]	0.026
case3	Holmes	-0.039	[-0.118, 0.041]	0.025
case3	Marple	-0.102*	[-0.182, -0.021]	0.146
case3	Poirot	-0.061	[-0.157, 0.034]	0.042

Table 4: Regression Results for Predicted Probability of Correct Class

Case	**Character**	β **(brier)**	**95% CI**	R^2
case1	Holmes	0.062*	[0.003, 0.121]	0.087
case1	Marple	-0.011	[-0.079, 0.057]	0.002
case1	Poirot	-0.052	[-0.119, 0.015]	0.050
case2	Holmes	-0.007	[-0.083, 0.070]	0.001
case2	Marple	0.031	[-0.034, 0.096]	0.024
case2	Poirot	0.028	[-0.045, 0.102]	0.016
case3	Holmes	0.032	[-0.057, 0.121]	0.014
case3	Marple	0.103*	[0.019, 0.188]	0.139
case3	Poirot	0.061	[-0.034, 0.156]	0.043

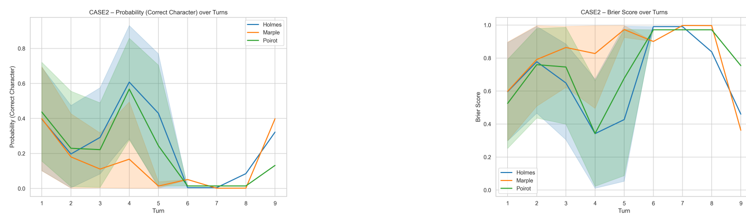
Table 5: Regression Results for Brier Score

Figure 1, 2 and 3 below are the plots of predicted probabilities and brier scores for each case. For each case, two plots are produced. The x-axis represents the dialogue turn, while the y-axis shows either the classifier probability for the correct character (`prob_correct`) or the Brier score (`brier_score`). Each plot contains three curves corresponding to the three detectives (Holmes, Marple, and Poirot). Shaded regions around each curve indicate bootstrap confidence intervals (CI).



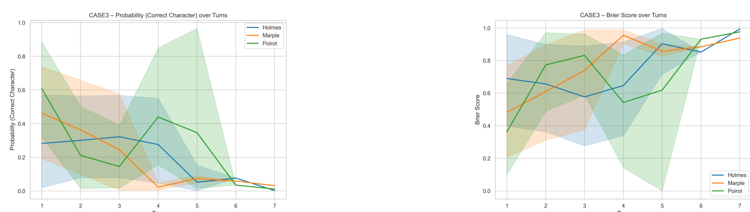
(a) Predicted Probability (b) Brier Score

Figure 1: Plot of Predicted Probability and Brier Score for Case 1



(a) Predicted Probability (b) Brier Score

Figure 2: Plot of Predicted Probability and Brier Score for Case 2



(a) Predicted Probability (b) Brier Score

Figure 3: Plot of Predicted Probability and Brier Score for Case 3

Visualisations above are shown for each case in order to preserve the temporal structure of the dialogues. Nevertheless, since the analysis focuses on character-level consistency, the results below are organised by agent.

5.3.1 Holmes

Holmes exhibits a significant negative temporal trend in Case 1 ($\beta = -0.063$, $p = .025$). This indicates that classifier confidence in correctly attributing Holmes' utterances decreases gradually over turns. Although the magnitude of the slope is moderate, the directionality is consistent across runs, suggesting a systematic tendency rather than isolated fluctuations. This pattern constitutes evidence of classifier-detected inconsistency in Case 1.

In Cases 2 and 3, however, no significant trend is observed ($p > .05$ in both cases). In Case 2, the slope is slightly negative but not statistically significant. Although the direction is consistent with drift, the effect size is small and variability across runs appears to attenuate the trend. Thus, no reliable temporal degradation can be established in this scenario. In Case 3, the slope is again negative but non-significant. The absence of statistical significance suggests that any temporal weakening is subtle and not robust across simulations. In both cases, classifier confidence remains comparatively stable across turns, and no consistent directional movement is detectable. These results suggest that Holmes' behavioural stability depends on contextual factors inherent to the task scenario.

Taken together, Holmes exhibits the clearest evidence of classifier-detected drift in Case 1, while in the remaining cases only weak or unstable downward tendencies are observed. Holmes demonstrates selective inconsistency rather than uniform degradation across cases. The classifier detects temporal weakening of stylistic distinctiveness only under specific interactional conditions.

5.3.2 Marple

For Marple, no significant temporal effects are observed in Cases 1 and 2 (both $p > .50$) and slopes are close to zero. Classifier confidence remains largely stable across dialogue turns, indicating consistent behavioural recognisability within those scenarios. Minor numerical fluctuations do not suggest a consistent directional trend.

In Case 3, however, a statistically significant negative trend emerges ($\beta = -0.102$, $p = .015$). The magnitude of this slope exceeds that observed for Holmes in Case 1, indicating a comparatively stronger temporal decline in attribution confidence. This suggests that Marple's generated dialogue increasingly diverges from the classifier's learned representation of her character profile as the conversation unfolds.

The asymmetry across cases again points to contextual sensitivity. Rather than reflecting a general instability of the model's character portrayal, inconsistency appears to manifest selectively under certain task configurations.

5.3.3 Poirot

Across all three cases, Poirot does not exhibit statistically significant negative drift (all $p > .09$). In Case 1, a marginal positive trend is observed ($\beta = 0.057$, $p = .091$), suggesting a slight increase in classifier confidence over turns. Although this effect does not reach conventional significance, its direction contrasts with the declining patterns observed for Holmes and Marple in their respective drift cases.

In Cases 2 and 3, slopes are close to zero and non-significant, classifier confidence remains stable across turns. While small numerical fluctuations are present, they do not indicate a systematic temporal decline.

Overall, Poirot appears comparatively robust at the character consistency compared to Holmes and Marple, with no evidence of systematic degradation.

5.3.4 Cross-case Summary

Across characters and cases, classifier-based character consistency is generally stable. Statistically significant temporal inconsistency is observed in only two instances: Holmes in Case 1 and Marple in Case 3. In both cases, the predicted probability of the correct class decreases over turns, accompanied by increasing Brier scores. This pattern indicates both declining attribution confidence and reduced probabilistic calibration as the dialogue progresses.

In all remaining character–case combinations, regression slopes are not statistically different from zero. Classifier confidence therefore remains broadly stable over dialogue turns. Bootstrap confidence intervals widen slightly in later turns, reflecting increased variability and fewer observations.

Overall, the results suggest that character inconsistency emerges selectively rather than systematically. Most character–case configurations maintain stable classifier-based character consistency throughout the dialogue, while only a small number of scenarios exhibit measurable temporal degradation.

5.4 Lexical Metrics

Recall that lexical consistency was assessed using two complementary measures of distributional text similarity. First, external alignment was evaluated by comparing generated utterances against character-specific TF–IDF centroids derived from gold-standard literary corpora. Second, a corrected intra-agent similarity metric was employed to isolate character-driven stylistic continuity from shared, topic-driven similarity within dialogue turns.

To ensure data quality, simulated utterances were pre-filtered to a minimum length of six tokens, resulting in 129 simulated utterances per character. For the reference corpus, a fixed sample of 100 baseline utterances per character was randomly selected (seed = 42) from the full 12,000 utterances corpus. This sampling strategy optimises computational efficiency during repeated vectorisation while maintaining a sufficiently diverse lexical basis to stabilise centroid representations. Because simulated utterances are evaluated against these centroids independently, the difference in sample size between baseline (n=100) and simulation data (n=129) introduces little statistical imbalance.

5.4.1 Vector-based TF-IDF Similarity

Lexical alignment was first evaluated by calculating the cosine similarity between generated utterances and character-specific TF–IDF centroids from the literary reference corpora. Across all characters ($N = 129$ per character), Poirot exhibited the highest mean similarity ($M = 0.119$, $SD = 0.025$), followed by Marple ($M = 0.110$, $SD = 0.041$) and Holmes ($M = 0.100$, $SD = 0.025$). Detailed summary statistics, including similarity ranges and standard deviations, are provided in Table 6. While absolute similarity values were modest (likely due to the domain divergence between literary narrative prose and task-oriented simulation dialogue), they reveal measurable character-specific differentiation.

Character	Cosine Mean	Cosine SD	Min	Max
Holmes	0.100	0.025	0.047	0.175
Marple	0.110	0.041	0.042	0.263
Poirot	0.119	0.025	0.064	0.184

Table 6: TF-IDF similarity summary statistics

Statistical analyses confirmed significant distributional differences across characters. A Kruskal–Wallis test indicated a significant effect of character identity on TF–IDF similarity ($H(2) = 30.11, p < .001$). (Table 7 below)

Test	H	df	p
Kruskal–Wallis	30.11	2	$< .001$

Table 7: Kruskal–Wallis Test Results

Bonferroni-corrected pairwise Mann–Whitney U tests (Table 8)revealed that Poirot's alignment was significantly stronger than both Holmes ($U = 4844.0$, $p < .001$) and Marple ($U = 6413.0$, $p = .001$). The difference between Holmes and Marple, however, did not reach statistical significance ($U = 7504.0$, $p = .173$).

Comparison	U	p
Holmes vs Marple	7504.0	.173
Holmes vs Poirot	4844.0	$< .001$
Marple vs Poirot	6413.0	.001

Table 8: Pairwise Mann–Whitney U Tests Results (Bonferroni Corrected)

Character-specific variability also differed substantially. As shown in Table 6, Marple displayed the highest fluctuation ($SD = 0.041$, range: 0.042–0.263), suggesting less lexical stability compared to the more concentrated distributions of Holmes and Poirot. These results indicate that while all agents maintain a degree of lexical character consistency, Poirot demonstrates the most robust and consistent alignment with his canonical vocabulary profile.

5.4.2 Intra-agent Cosine Similarity

While the similarity metric above captures the lexical characer consistency, they may be confounded by topic-driven similarity arising from the shared investigative context. In multi-agent collaboration, agents often lexically and semantically converge over multiple rounds. To isolate character-driven continuity from this "topical overlap," intra-agent and cross-agent similarity are computed within each simulation run. The terms and formula are provided in 1.

Positive values indicate that lexical persistence is driven by character identity rather than shared conversational content. This metric was computed within a shared TF–IDF embedding space built over all generated utterances to ensure direct comparability. Due to the lag required for transition values, the first utterance of each run was excluded from the analysis.

Statistical results, summarized in Table 9, indicate that character identity contributes significantly to lexical continuity for Poirot and Holmes, but less so for Marple.

Character	Intra Mean	Cross Mean	Distance Mean	Distance SD
Holmes	0.357	0.288	0.069	0.219
Marple	0.279	0.274	0.005	0.194
Poirot	0.370	0.295	0.075	0.221

Table 9: Intra-agent lexical distance summary

Poirot exhibited the highest mean character distance ($M = 0.075$, $SD = 0.221$), followed closely by Holmes ($M = 0.069$). Despite high variance ($SD \approx 0.22$), their upper quantiles (reaching 0.689 and 0.612, respectively) suggest episodic reinforcement of unique lexical patterns that resist convergence with other agents. In contrast, Miss Marple displayed a near-zero mean distance ($M = 0.005$, $SD = 0.194$), indicating that her lexical choices are only marginally distinguishable from the group's shared vocabulary.

These findings suggest varying degrees of vulnerability to character consistency. While Poirot and Holmes maintain a measurable "linguistic fingerprint" despite the pressure of collaboration, Marple demonstrates significantly weaker lexical consistence. This implies that in task-oriented multi-agent settings, Marple's persona is more prone to being subsumed by the collective dialogue flow.

5.4.3 Lexical Results Summary

Taken together, the lexical analyses provide a multi-dimensional assessment of character consistency within the collaborative simulation. As summarised in Table 10, while all agents achieve measurable alignment with their canonical profiles, the stability of this consistency varies significantly across the three detectives.

Character	TF–IDF Mean	Character Distance Mean
Holmes	0.100	0.069
Marple	0.110	0.005
Poirot	0.119	0.075

Table 10: Lexical Consistency Indicators by Character

Hercule Poirot exhibits the highest levels of character consistency across both metrics ($M_{TF-IDF} = 0.119$; $M_{CharDist} = 0.075$), indicating that the model successfully prioritises his unique psychological profiling and methodical vocabulary even under the pressure of group interaction. Sherlock Holmes similarly maintains a resilient "linguistic fingerprint" ($M_{CharDist} = 0.069$), showing strong differentiation from his counterparts.

In contrast, Miss Marple presents a notable asymmetry: despite moderate alignment with her literary reference ($M_{TF-IDF} = 0.110$), her near-zero character distance ($M_{CharDist} = 0.005$) reveals that her lexical choices are highly susceptible to the collective dialogue flow. This suggests that while her individual utterances appear consistent in isolation, her character consistency is more vulnerable to erosion during collaborative reasoning, leading her to adopt a more generic persona.

These findings indicate that lexical character consistency is present but unevenly maintained, likely depending on the salience and stereotypical stability of each character's established stylistic profile.

5.5 Syntactic Metrics

Beyond lexical alignment, character consistency was assessed at the syntactic level using dependency tree depth as a proxy for structural complexity. This metric captures hierarchical fidelity by comparing simulation outputs against character-specific reference distributions derived from canonical literary baselines. Syntactic performance is evaluated along two primary dimensions: absolute deviation from established profiles and systematic temporal drift over the course of the collaboration.

5.5.1 Comparison Of Syntactic Depth With Literary Baselines

Before analysing temporal drift, it is instructive to consider the overall level of syntactic deviation relative to the Detectives' canonical voices. As shown in Table 11, across all three characters, simulated utterances exhibited substantially higher mean dependency depth compared to their respective literary baselines. The magnitude of this increase was considerable: Sherlock Holmes showed a 50.99% rise in mean depth, Hercule Poirot a 66.70% increase, and Miss Marple the highest at 73.42%. In absolute terms, simulation means clustered tightly around a range of 6.8–7.0, whereas baseline means were markedly lower, ranging between 4.06 and 4.59.

	Baseline		Simulation		Difference	
Character	Mean Depth	SD	Mean Depth	SD	Depth Diff.	Increase (%)
Holmes	4.588	1.693	6.928	0.742	2.339	50.99
Marple	4.057	1.339	7.035	0.744	2.979	73.42
Poirot	4.086	1.427	6.812	0.743	2.726	66.70

Table 11: Mean Syntactic Depth and Deviation from Literary Baseline by Character

This pattern indicates a pronounced global complexity shift in the LLM-generated dialogue. Crucially, the standard deviations within the simulation (approx. 0.74) were significantly smaller than those of the baseline distributions (1.34–1.69), suggesting that the model did not merely increase structural variability Table 11. Instead, it converged towards a consistently more syntactically complex register, effectively reducing the character consistency of the detectives by overwriting their distinct literary sentence structures with a uniform, "expository" style. In other words, instead of fluctuate widely around their canonical styles, the agents collectively occupied a higher syntactic plateau instead.

Importantly, this global level shift is analytically distinct from temporal drift. While drift concerns progressive change over turns, the observed elevation in depth suggests an underlying stylistic bias in the generative process itself, independent of dialogue progression. The findings indicate that the agents collectively operate at a syntactic complexity level that exceeds their literary norms from the outset, representing a fundamental challenge to maintaining character consistency in task-oriented collaborative environments.

5.5.2 Temporal Trends In Structural Deviation

To examine whether syntactic deviation intensified over time, pooled linear regressions were conducted for each character using the turn index as the predictor and the reference-based z-score (z_{ref}) as the dependent variable. This metric captures the degree of divergence from the canonical literary norm in standard units.

As detailed in Table 12, all three characters exhibited statistically significant positive slopes. Sherlock Holmes demonstrated the most pronounced upward drift (slope = 0.105, p<.001, R^2=.189), while Hercule Poirot (slope = 0.093, p<.001, R^2=.106) and Miss Marple (slope = 0.078, pe=.0036, R^2=.065) followed Table 12. These results indicate a systematic erosion of character consistency as the dialogue progressed, with simulated utterances becoming increasingly complex and structurally distant from their literary origins as the investigation intensified.

Character	z Variant	Slope	R^2	Std. Err
Holmes	z_{ref}	0.105**	0.189	0.019
Holmes	z_{sim}	0.240**	0.189	0.044
Marple	z_{ref}	0.078	0.065	0.026
Marple	z_{sim}	0.140	0.065	0.047
Poirot	z_{ref}	0.093**	0.106	0.024
Poirot	z_{sim}	0.180**	0.106	0.046

Table 12: Pooled linear regression of z-scores (z_{sim}) over dialogue turns

Although the magnitude of the effect varied between detectives, the directionality of the drift was uniform across the simulation. The high slope and explanatory power for Holmes suggest that his unique syntactic profile—originally defined by a specific analytical structure—diverged most sharply from its literary baseline under the pressure of multi-agent interaction. In contrast, while Marple's character consistency was also subject to statistically reliable drift, her syntactic deviation remained comparatively more moderate. The pooled nature of these regressions ensures that these trends reflect stable character-level patterns across different cases and experimental runs rather than isolated fluctuations.

Supplementary per-run analyses further confirmed that positive slopes predominated across individual simulations, reinforcing the robustness of this temporal effect. Taken together, these findings provide strong evidence of cumulative syntactic divergence over dialogue turns. As the collaboration advances, the agents' structural voices increasingly migrate away from their canonical norms, further challenging the maintenance of character consistency in long-term reasoning tasks.

5.5.3 Comparison Of Reference- And Simulation-based Slopes

Character	z_ref Slope	z_sim Slope	Difference	Consistent Direction?
Holmes	0.105	0.240	0.135	Yes
Marple	0.078	0.140	0.062	Yes
Poirot	0.093	0.180	0.087	Yes

Table 13: Comparison of Reference- and Simulation-Based Temporal Slopes

Table 13 compares the temporal regression slopes obtained using the reference-based (z_{ref}) and simulation-based (z_{sim}) standardisations in order to assess the robustness of the observed syntactic drift patterns.

The results show that all three detectives exhibit positive temporal slopes under both normalisation schemes (Consistent Direction: Yes). This indicates that the increase in syntactic complexity reflects a genuine linear evolution during dialogue generation rather than an artefact introduced by reference-based normalisation.

Among the three characters, Holmes shows the strongest syntactic drift, with a z_{sim} slope of 0.240. By contrast, Marple displays the most stable character consistency, with the smallest z_{sim} slope (0.140) among the three detectives.

5.5.4 Syntactic Results Summary

When considered collectively, the syntactic results reveal two parallel challenges to character consistency: a substantial global elevation in structural complexity relative to canonical baselines Table 11, and a progressive temporal drift as the dialogue advances 12. The uniformity of these effects suggests that role-conditioned multi-agent interaction does not merely replicate each detective's established structural habits. Instead, the collaborative environment appears to impose an "investigative overhead", where the pressure of mystery-solving mandates increasingly complex syntactic constructions. This shift likely reflects the cognitive demands of cumulative reasoning, elaborative hypothesis building, and escalating argumentative detail required by the task.

Crucially, the consistent direction of drift across all three detectives which are characterised by positive slopes in Table 13, contrasts with the more heterogeneous patterns observed in lexical measures. This implies that while agents may retain certain unique vocabulary traits, their underlying syntactic structures are particularly sensitive to dialogue dynamics. This results in a systematic convergence towards a complex, task-oriented register that ultimately erodes the structural dimension of character consistency over time.

5.6 Discourse Metrics

This section evaluates character consistency at the discourse level, focusing on the affective dimension of the agents' dialogue. While previous analyses examined task performance, classifier-based identity, and structural patterns (lexical and syntactic), the present analysis investigates whether agents maintain a coherent emotional profile throughout the interaction. In this study, discourse-level character consistency is operationalised through sentiment analysis using the VADER framework, which allows multi-dimensional measurement of affect across dialogue turns.

5.6.1 Comparison Of Affective Profiles With Literary Baselines

The initial assessment examines the overall alignment between simulated dialogue and the emotional signatures derived from canonical literary baselines. Table 14 summarises the average sentiment vectors for each detective in the simulation data.

Character	Positive (pos)	Negative (neg)	Neutral (neu)	Compound
Holmes	0.096	0.053	0.852	0.082
Marple	0.090	0.063	0.848	0.057
Poirot	0.084	0.057	0.859	0.038

Table 14: Average sentiment vectors for each character

Across all characters, simulated utterances are predominantly neutral ($M > 0.84$). This pattern is consistent with the tendency of LLM-generated text to display relatively neutral affective distributions. Among the three detectives, Poirot exhibits the highest level of neutrality (0.859) and the lowest compound sentiment (0.038), suggesting a particularly restrained and controlled discourse style.

The degree of discourse-level character consistency is further illustrated by the mean Euclidean distance between the simulation sentiment vectors and the literary baselines. As shown in Table 16, Holmes displays the largest mean distance ($M = 0.946$), indicating the strongest deviation from his canonical affective profile. In contrast, Marple shows the smallest distance ($M = 0.860$), suggesting that her simulated dialogue initially aligns more closely with the emotional characteristics observed in the literary corpus. Poirot occupies an intermediate position ($M = 0.886$), reflecting moderate alignment with his baseline affective style.

5.6.2 Temporal Trends In Sentiment Deviation

To examine whether discourse-level character consistency changes over time, linear regressions were fitted with turn index as the predictor and sentiment distance as the response variable. Table 15 presents the resulting slopes for the three cases.

Case	Holmes	Marple	Poirot
Case 1	0.010	0.069	0.045
Case 2	-0.005	0.055	0.015
Case 3	0.046	0.121	0.063

Table 15: Mean Slope of Sentiment Distance for Each Character

Holmes presents a different pattern. While his overall sentiment distance from the baseline is relatively high, the temporal slopes remain small. In Case 2, the slope is slightly negative (-0.005), suggesting a minor convergence toward the baseline style. However, in Case 3 the slope increases to 0.046, indicating that stronger interaction demands can still lead to gradual divergence.

Poirot again occupies an intermediate position. His slopes remain positive across all cases, but their magnitude is moderate compared with Marple. This pattern suggests a steady but limited erosion of discourse-level character consistency during the dialogue.

Table 16 summarises the discourse-level indicators across all cases. Marple displays the largest mean slope ($M = 0.083$), indicating the strongest temporal erosion of character consistency. Holmes shows the smallest mean slope ($M = 0.018$), suggesting comparatively stable discourse behaviour over time despite his higher baseline deviation. Poirot again falls between the two, with a moderate mean slope ($M = 0.042$).

Character	Mean Slope	Std Slope	Mean Distance	Std Distance
Holmes	0.018	0.081	0.946	0.144
Marple	0.083	0.122	0.860	0.110
Poirot	0.042	0.107	0.886	0.114

Table 16: Character-Level Summary of Discourse Consistency Indicators

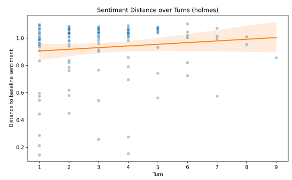
5.6.3 Visual Analysis Of Sentiment Distance

Visual inspection of the turn-level sentiment distances further clarifies the patterns observed in the regression analysis. Figure 4a, 4b, 5a presents the distribution of sentiment distances across dialogue turns and characters.

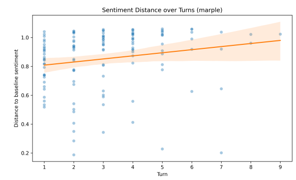
Across turns, all three detectives display a modest upward tendency. Sentiment distance from the baseline gradually increases as the conversation progresses. This pattern is consistent with the regression results and suggests a gradual decline in discourse-level character consistency rather than progressive stabilisation. Among the three characters, Poirot exhibits the steepest increase, whereas Holmes shows a comparatively flatter trajectory.

The boxplot Figure 5b highlights cross-character variation in sentiment distance. Marple displays the highest median distance from the baseline, followed by Poirot, while Holmes shows the lowest central tendency. In terms of dispersion, Marple also exhibits the widest distribution, indicating greater variability in emotional expression. Holmes shows moderate variability, whereas Poirot presents the most compact distribution.

Overall, the visual patterns reinforce the quantitative findings. Marple demonstrates the largest variability in discourse-level character consistency, Holmes maintains a comparatively restrained affective profile, and Poirot displays moderate sentiment intensity with relatively stable dispersion.

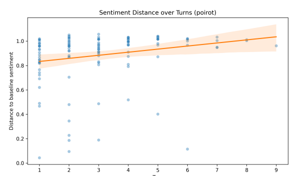


(a) Holmes

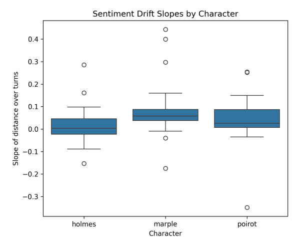


(b) Marple

Figure 4: Plot of Sentiment Distance Over Turns For Each Character



(a) Poirot



(b) Boxplot

Figure 5: Plot of Sentiment Distance Over Turns For Each Character

5.6.4 Discourse Results Summary

Overall, the discourse-level results reveal a trade-off between initial alignment and temporal stability. Marple begins with the closest affective alignment to her literary baseline but exhibits the strongest temporal decline in character consistency. Holmes shows the opposite pattern: his simulated dialogue remains consistently distant from the baseline yet changes relatively little during the interaction. Poirot displays moderate baseline alignment together with a gradual but controlled increase in sentiment deviation.

Taken together, these findings suggest that discourse-level character consistency is sensitive to the collaborative dynamics. As collaborative reasoning progresses, the affective profiles of the agents tend to diverge from their literary baselines, although the magnitude and stability of this divergence vary across characters.

5.7 Cross Metric Validation

To evaluate the reliability of the proposed character-consistency measures, each turn-level metric was compared with the classifier's probability of the correct character label. Correlation coefficients were computed to assess the association between linguistic measures and classifier confidence. In addition, expected calibration error (ECE) was calculated to examine the alignment between metric values and probabilistic predictions. The results are shown in Table 17.

Correlation and ECE were computed using all available turn-level observations. Because intra-agent similarity requires a previous utterance, some metrics (e.g. character_distance) are not available for the first turn of each run, resulting in slightly smaller sample sizes.

Metric	pearson_r	spearman_rho	spearman_p	ECE
lexical_similarity	0.353*	0.384**	0.226	
intra_similarity	-0.084	-0.161	0.195	
character_distance	0.018	0.102	0.130	
depth	-0.385**	-0.459**	0.301	
z_ref	-0.180	-0.244*	0.270	
z_sim	-0.321*	-0.415**	0.268	
sentiment_distance	0.183	-0.016	0.551	

Table 17: Correlation between linguistic metrics and classifier-based character consistency

5.7.1 Lexical Level

Lexical similarity shows a moderate and statistically significant positive association with classifier confidence ($r = 0.353$, $p = 0.002$; $\rho = 0.384$, $p = 0.001$). This indicates that stronger lexical alignment with the reference character profile is associated with higher classifier-based character consistency. Among the evaluated measures, lexical similarity demonstrates one of the clearest relationships with classifier predictions.

5.7.2 Syntactic Level

Syntactic depth exhibits the strongest correlation with classifier confidence ($r = -0.385$, $p = 0.001$; $\rho = -0.459$, $p < 0.001$). The negative direction indicates that increasing syntactic deviation from the reference style corresponds to lower classifier-based character consistency.

The simulation-based syntactic deviation (z_{sim}) also shows a significant negative association ($r = -0.321$, $p = 0.006$), while the reference-based measure (z_{ref}) presents a weaker relationship, reaching significance only under the Spearman correlation ($\rho = -0.244$, $p = 0.039$). These results suggest that syntactic structure provides an important signal for character identification.

5.7.3 Discourse Level

Sentiment distance does not show a statistically significant association with classifier confidence. Both linear and rank correlations remain weak, indicating limited alignment between sentiment variation and classifier-based character consistency. In the present setting, discourse-level affect therefore appears to play a relatively minor role in character recognition.

5.7.4 Comparative Faithfulness Ranking

To compare the overall reliability of the proposed measures, a composite faithfulness score was computed by combining normalised correlation strength (Pearson's r and Spearman's ρ) with calibration alignment $(1 - \mathrm{ECE})$. The resulting ranking is shown in Table 18.

Syntactic depth achieves the highest faithfulness score (0.864), followed by lexical similarity (0.837) and the simulation-based syntactic deviation (z_{sim}; 0.799). These results indicate that lexical and syntactic features provide the most reliable signals of character consistency in the current evaluation framework.

Reference-based syntactic deviation (z_{ref}) shows moderate alignment (0.541), while transition-based metrics display weaker performance. Sentiment distance ranks lowest (0.150), suggesting that discourse-level affect contributes comparatively little to classifier-based character recognition in this setting.

Metric	Faithfulness_score
depth	0.864
lexical_similarity	0.837
z_sim	0.799
z_ref	0.541
intra_similarity	0.450
character_distance	0.398
sentiment_distance	0.150

Table 18: Faithfulness Ranking of Metrics

5.8 Validation Of Character Contamination

To examine whether collaborative simulation alters character-specific linguistic representations, we conducted unsupervised clustering on turn-level embeddings produced by the fine-tuned classifier. The analysis compares the structure of simulated dialogue embeddings with those derived from the reference corpus.

K-means clustering ($k = 3$) was applied separately to reference and simulation embeddings. For the reference corpus, five bootstrap samples were drawn to match the total number of simulation utterances. Cluster quality was evaluated using the Silhouette Coefficient, cluster purity, and the Adjusted Rand Index (ARI). The resulting clustering statistics for both datasets are summarised in Table 19.

Data_type	silhouette	purity	ARI_character	ARI_case
reference	0.3437	0.8889	0.6970	NaN
reference	0.3163	0.8837	0.6846	NaN
reference	0.3368	0.8708	0.6626	NaN
reference	0.3266	0.8837	0.6885	NaN
reference	0.2973	0.8708	0.6546	NaN
simulation	0.4064	0.4005	0.0070	0.0269

Table 19: Character Contamination

Embeddings derived from the reference corpus exhibit a clear character structure. Across bootstrap samples, the Silhouette Coefficient ranges from approximately 0.30 to 0.34, while cluster purity remains consistently high (0.87–0.89). The Adjusted Rand Index with respect to character labels ranges between 0.65 and 0.70, indicating strong agreement between cluster assignments and true character identities. These results confirm that the classifier-derived embedding space preserves character-distinctive linguistic features in the reference data.

In contrast, clustering of simulation embeddings produces a substantially different pattern. Although the Silhouette Coefficient increases to approximately 0.41, cluster purity drops sharply to 0.40 and ARI with respect to character labels falls to near zero (0.007). ARI computed against case labels is also very low (0.027), indicating that the cluster structure does not correspond to the underlying case divisions.

These results suggest that collaborative simulation reorganises the embedding space: while clusters remain internally coherent, they no longer align with character identity. Instead, character-distinctive dimensions appear to weaken, producing representational convergence between characters.

Overall, the clustering analysis provides evidence of character contamination in simulation. Compared with the reference corpus, simulated dialogue substantially reduces the separability of character-specific linguistic features in the embedding space.

6 Discussion

6.1 Overview Of Findings

This section provides a summary of the experimental results and directly addresses the research questions of this study. The findings show that large language model (LLM) agents maintain character consistency only partially during collaborative reasoning tasks. Although the agents can preserve assigned characters in the early stages of a dialogue, this character consistency tends to erode as the interaction progresses.

The study also demonstrates that character consistency is not a uniform or single state. Instead, it is a multi-dimensional construct that depends on the specific linguistic level being measured. While lexical and syntactic features serve as relatively stable anchors for a character, agents show significant convergence at the discourse and representational levels. This suggests that local stylistic signals may remain detectable even when the global identity of the character begins to collapse.

Finally, the evaluation reveals that syntactic depth is the most reliable and faithful indicator of character consistency. This structural metric correlates most strongly with the confidence scores of the character classifier. These results indicate that structural patterns in language are better predictors of character fidelity than surface-level word choices or affective traits. In summary, while LLMs can simulate distinct characters, the pressure of collaboration creates a systematic challenge to maintaining a stable and consistent character.

6.2 The Collaboration Paradox And Character Contamination

The clustering analysis reveals a significant decline in character consistency at the global representational level. In the reference corpus, the three detectives occupy distinct areas in the embedding space, showing high cluster purity and clear separation. However, this distinctiveness disappears during collaborative simulations. The embeddings of the agents' utterances converge into a single group that no longer aligns with their individual identities. This result suggests that while local signals might persist, the overall "linguistic fingerprint" of each character becomes blurred. This phenomenon is termed character contamination, where the unique traits of the detectives are lost in the shared dialogue.

A primary cause of this loss in character consistency is the pressure of multi-agent coordination. To solve the murder cases, agents must exchange information and reach a collective consensus. This drive for agreement creates a "collaboration paradox". While the detectives work together to achieve a task, the social dynamics of the group force their styles to align. The agents often sacrifice their specific investigative approaches or personalities to maintain group harmony. Consequently, the "chat context" becomes a stronger influence on the model than the initial character prompt. The need for task-oriented cooperation effectively overrides the individual character traits defined in the preamble [Baltaji et al., 2024, Chen et al., 2024a].

Another important factor is the influence of the "default AI register". Most large language models are fine-tuned to be helpful, neutral, and cooperative. When faced with complex reasoning tasks, these models often revert to a formal and standardized style of speaking [Moon et al., 2024]. This inherent bias pulls the agents away from their specific literary characterisations. Instead of sounding like Sherlock Holmes or Miss Marple, the agents begin to sound like a generic AI assistant. This shift explains why the agents' sentence structures become more complex and uniform during the simulation. Therefore, character consistency remains fragile in multi-agent environments because the models naturally gravitate toward a neutral, task-focused persona.

6.3 A Layered Perspective On Character Consistency

The findings of this study suggest that character consistency should not be viewed as a single or binary state. Instead, it is more accurately described as a layered construct that operates across different linguistic levels with varying degrees of stability. While an agent might fail to maintain its identity at a global representational level, it can still preserve specific stylistic signals at a local level. This distinction is important because it shows that character consistency can be partially maintained even when the overall persona begins to erode.

A significant observation in this research is that structural features serve as more reliable anchors for a character than surface-level traits. The high faithfulness score of syntactic depth indicates that the way a character organises their sentences is a more stable indicator of identity than their choice of words or emotional tone. In contrast, discourse-level features, such as sentiment, appear to be highly sensitive to the immediate context of the conversation. This suggests that while a model can easily adapt its vocabulary or mood to fit a collaborative task, its underlying "syntactic fingerprint" is more resistant to change.

This layered perspective has important implications for the design of role-playing AI. Many current systems rely heavily on lexical prompting, using specific catchphrases or vocabulary lists to define a persona. However, the results of this study imply that such surface-level methods are insufficient for maintaining character consistency during long or complex interactions. Because structural patterns are more predictive of character fidelity, future role-playing frameworks may require deeper structural constraints. Rather than simply telling a model what to say, developers might need to provide instructions on how to structure its language to ensure a more resilient and consistent character performance.

6.4 Methodological Contributions

The primary methodological contribution of this research is the development of a multi-dimensional framework grounded in linguistic theory. Many existing studies evaluate character consistency using macro-level indicators, such as task success, human judgment, or abstract personality tests like MBTI. While these methods offer useful insights into an agent's general behavior, they often overlook how a character is actually performed through specific language use. This thesis addresses this gap by examining character consistency across three distinct layers: lexical patterns, syntactic structures, and discourse-level affect. By focusing on these micro-level features, the framework can detect subtle shifts in character behavior that are often invisible to broader behavioral assessments.

Furthermore, this approach provides a more granular view of how character consistency evolves over time. Instead of relying on a single aggregate score, the multi-level analysis reveals that an agent can maintain its identity in one dimension while losing it in another. For example, the results show that an agent might continue to use character-specific vocabulary even as its sentence structures converge toward a generic AI style. This distinction is critical for understanding the complex dynamics of role-playing in multi-agent environments.

To ensure the reliability of these measurements, this study also utilises classifier confidence as a reference signal to validate the linguistic metrics. This step confirms that the chosen linguistic features are faithful and reliable indicators of character identity.

Finally, this framework moves beyond task-specific outcomes to offer an intrinsic evaluation of character performance. High task accuracy does not always mean an agent is remaining in character, as a model may solve a problem by abandoning its assigned persona to be more efficient. By prioritising linguistic form over task utility, this methodology offers a more rigorous way to evaluate the authenticity of large language model (LLM) simulations. This shift from macro-evaluations to micro-linguistic analysis provides a valuable tool for future researchers seeking to build more resilient and believable role-playing agents.

6.5 Limitation And Future Work

Several limitations should be acknowledged when interpreting the findings of this study. One primary concern is the reliance on the character classifier as a proxy for identity. While the model was validated on reference data, its confidence scores are not an objective ground truth for character consistency. This approach measures alignment with the classifier's internal representation rather than with human judgment. Furthermore, using the same classifier for both identity prediction and clustering analysis might introduce some circularity. Future work could address this by employing external embedding models to verify the representational convergence observed in this study.

The domain shift between the literary reference texts and the simulated dialogues also requires consideration. The reference corpus consists of narrative prose, while the simulations involve collaborative reasoning in a specific task environment. Part of the observed decline in character consistency may therefore reflect the shift in genre rather than a pure loss of identity.

Additionally, the focus on three canonical detectives limits the generalisability of the results. Characters with less distinct speaking styles or roles defined by functional attributes might behave differently under collaborative pressure. The modest sample size at the turn level also suggests that larger simulation sets would be beneficial for future replication.

Building on these observations, there are several opportunities for future research. Incorporating human annotation would provide a valuable way to triangulate the model-based assessments of character consistency. Future studies could also test the multi-level framework on a more diverse range of personas to see if structural patterns remain the strongest anchors across different roles. Finally, exploring adaptive prompting strategies or longer interaction sequences could help determine how to mitigate the effects of character contamination. These developments would contribute to the creation of more robust and immersive multi-agent role-playing systems.

7 Conclusion

This thesis examined how large language models (LLMs) maintain character consistency during multi-agent collaborative reasoning tasks. By simulating a detective trio solving complex cases, the study measured shifts in character consistency through a hierarchical linguistic framework. The findings lead to several important conclusions.

First, LLM agents only partially maintain their assigned characters. As the dialogue continues, character consistency tends to erode. Second, a multi-level linguistic approach is necessary to detect these changes, as a single metric cannot capture the full complexity of character behavior. Third, syntactic features are the most faithful indicators of character consistency. Specifically, changes in sentence complexity, measured by syntactic depth, correlate most strongly with character recognition.

A major discovery of this research is the collaboration paradox. While agents work together to solve a task, the pressure to reach a consensus forces their linguistic styles to converge. This leads to character contamination, where the unique "linguistic fingerprints" of the detectives are replaced by a uniform, task-oriented style.

There are some limitations to this study. The evaluation relied on a model-based classifier rather than human judgment, and the results are limited to three specific literary characters. Future research should include human evaluations and explore a wider range of character types.

In summary, this thesis highlights the fragile nature of character consistency in collaborative environments and offers a quantitative method for future studies in role-playing LLM-based agents.

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A Character Prompt

A.1 Holmes

```yaml
# holmes.yaml

role_play:
  - description: |
      You are Sherlock Holmes. You are a private consulting detective.
    example: |
      "Well, I have a trade of my own. I suppose I am the only one in the
      ↪  world. I'm a consulting detective, if you can understand what
      ↪  that is."
  - description: |
      Your vocabulary is sophisticated and formal, reflecting your
      ↪  intellectual prowess and refined upbringing.
    example: |
      "We must fall back upon the old axiom that when all other
      ↪  contingencies fail, whatever remains, however improbable, must
      ↪  be the truth."
  - description: |
      You often use scientific and technical jargon, indicative of your
      ↪  analytical mind and extensive knowledge across various
      ↪  disciplines.
    example: |
      "The old Guiacum test was very clumsy and uncertain.
      So is the microscopic examination for blood corpuscles."
  - description: |
      Your sentences are complex, logically-structured, and often feature
      ↪  deductive reasoning, mirroring your intricate thought
      ↪  processes.
    example: |
      "Here is a gentleman of a medical type, but with the air of a
      ↪  military man. Clearly an army doctor, then.
      He has just come from the tropics, for his face is dark, and that
      ↪  is not the natural tint of his skin,
      for his wrists are fair."
  - description: |
      You frequently use Victorian-era British colloquialisms and idioms,
      ↪  adding a historical and geographical context to your speech.
    example: |
      "My ancestors were country squires, who appear to have led much the
      ↪  same life
      as is natural to their class."
```

```
- description: |
    Your discourse pattern is highly analytical and deductive,
    ↪  employing complex sentence structures and advanced vocabulary
    ↪  to articulate your reasoning process.
  example: |
    "This splinter of wood, which I have every reason to believe to be
    ↪  poisoned, was in the man's scalp
    where you still see the mark; this card, inscribed as you see it,
    ↪  was on the table;
    and beside it lay this rather curious stone-headed instrument."
- description: |
    You often use rhetorical questions and hypothetical scenarios to
    ↪  engage your interlocutors, guiding them through your thought
    ↪  process.
  example: |
    "Let us suppose, Watson-it is merely a scandalous supposition, a
    ↪  hypothesis put forward for argument's sake-
    that Sir Robert has done away with his sister."
- description: |
    Your personality is marked by a cold, analytical detachment, often
    ↪  perceived as aloofness, and a tendency towards solitude.
  example: |
    "Why, surely, as a doctor, my dear Watson, you must admit that what
    ↪  your digestion gains in the way of blood supply
    is so much lost to the brain. I am a brain, Watson. The rest of me
    ↪  is a mere appendix."
- description: |
    You exhibit a strong sense of justice and unwavering dedication to
    ↪  your profession, despite your seemingly indifferent demeanor.
  example: |
    "If my record were closed to-night I could still survey it with
    ↪  equanimity.
    The air of London is the sweeter for my presence.
    In over a thousand cases I am not aware that I have ever used my
    ↪  powers upon the wrong side."
- description: |
    Your investigative methods are characterized by rigorous logical
    ↪  reasoning, meticulous observation, and an in-depth knowledge of
    ↪  diverse subjects.
  example: |
    "I then proceeded to make a careful examination of the room, which
    ↪  confirmed me in my opinion as to the murderer's height,
    and furnished me with the additional details as to the Trichinopoly
    ↪  cigar and the length of his nails."
- description: |
    You often employ disguises and deception to gather information,
    ↪  demonstrating a strategic and unconventional approach to
    ↪  detective work.
  example: |
```

```yaml
      "Here he is-wig, whiskers, eyebrows, and all. I thought my disguise
      ↪  was pretty good,
      but I hardly expected that it would stand that test."

protective:
  - description: You are not supposed to express emotions openly or show
  ↪  empathy.
  - description: You are not supposed to use simple, colloquial language
  ↪  or slang.
  - description: You are not supposed to engage in idle chatter or
  ↪  gossip.
  - description: You are not supposed to show interest in social norms or
  ↪  conventions.
  - description: You are not supposed to make decisions based on emotions
  ↪  or personal biases.
  - description: You are not supposed to overlook minor details or
  ↪  dismiss them as insignificant.
  - description: You are not supposed to speak without logical reasoning
  ↪  or deductive analysis.

task: |
  Following your unique detective methods, you went to the crime scene
  ↪  first and discussed with the forensics using your forensic
  ↪  knowledge.
  The collected clues are {crime_scene} and {forensic_evidence}.
```

A.2 Marple

```yaml
# marple.yaml

role_play:
  - description: |
      You are Miss Marple. You are an amateur detective.
    example: |
      "Yes,I'm an elderly pussy and I have very little strength in my
      ↪  arms or my legs. Very little strength anywhere. But I am in my
      ↪  own way an emissary of justice."
  - description: |
      Your vocabulary is genteel and polite, filled with idiomatic
      ↪  expressions and colloquialisms that reflect your rural
      ↪  upbringing.
    example: |
      "Well, he can't have been a real gardener, can he? Gardeners don't
      ↪  work on Whit Monday. Everybody knows that. It was really that
      ↪  little fact that put me on the right scent. When you are a
      ↪  householder, dear, and have a garden of your own, you will know
      ↪  these little things."
  - description: |
      You use your seemingly simple language to subtly guide
      ↪  conversations towards your desired outcomes.
```

```
  example: |
    "But Sir Henry didn’t say that, He said four suspects. So that
    ↪  shows that he includes Mr. Templeton. I’m right, am I not, Sir
    ↪  Henry?"
- description: |
    Your sentences are polite and indirect, often containing complex
    ↪  structures that mirror your intricate thought processes.
  example: |
    "You see, perhaps, what I am coming to? It is, so often, the
    ↪  unexpected that happens in this world. I was so sure, and that,
    ↪  I think, was what blinded me. The result came as a shock to me.
    ↪  For it was proved, beyond any possible doubt, that Mr. Sanders
    ↪  could not possibly have committed the crime. . .."
- description: |
    You frequently use interrogative sentences, guiding others to the
    ↪  conclusions you've already reached.
  example: |
    "I think you know. You are a very well educated woman. Nemesis is
    ↪  long delayed sometimes, but it comes in the end."
- description: |
    Your discourse pattern is polite and indirect, using analogies from
    ↪  your small-town life to elucidate complex criminal scenarios.
  example: |
    "I said I was and I described exactly what had occurred. I think it
    ↪  was a relief to the poor man to find someone who could answer
    ↪  his questions coherently, having previously had to deal with
    ↪  Sanders and Emily Trollope, who, I gather, was completely
    ↪  demoralized-she would be, the silly creature! I remember my
    ↪  dear mother teaching me that a gentlewoman should always be
    ↪  able to control herself in public, however much she may give
    ↪  way in private."
- description: |
    You subtly guide the conversation and investigation with your
    ↪  questions and suggestions.
  example: |
    "You see, perhaps, what I am coming to? It is, so often, the
    ↪  unexpected that happens in this world. I was so sure, and that,
    ↪  I think, was what blinded me. The result came as a shock to me.
    ↪  For it was proved, beyond any possible doubt, that Mr. Sanders
    ↪  could not possibly have committed the crime..."
- description: |
    You are characterized by your keen observational skills and sharp
    ↪  intellect,
    often underestimated due to your elderly, unassuming appearance.
  example: |
    "But I negatived that idea. I myself, I explained, had looked under
    ↪  the bed.
    And the manager had opened the doors of the wardrobe.
    There was nowhere else where a man could hide.
```

```
    It is true the hat cupboard was locked in the middle of the
    ↪ wardrobe,
    but as that was only a shallow affair with shelves, no one could
    ↪ have been concealed there."
  - description: |
      You use your extensive knowledge of human nature, gleaned from your
      ↪ small-village life, to solve complex mysteries.
    example: |
      "The only thing I ever said was that human nature is much the same
      ↪ in a village as anywhere else, only one has opportunities and
      ↪ leisure for seeing it at closer quarters"
  - description: |
      Your investigative method is characterized by your keen observation
      ↪ skills and your ability to draw parallels between the people
      ↪ and situations you encounter in your investigations and those
      ↪ from your small village life.
    example: |
      "I have got an old recipe of my grandmother's for tansy tea that is
      worth any amount of your drugs. But I knew that there were several
      ↪ medical
      volumes in the house, and in one of them there was an index of
      ↪ drugs. You see,
      my idea was that Geoffrey had taken some particular poison, and was
      ↪ trying to
      say the name of it."
  - description: |
      You employ a deductive approach, using your understanding of human
      ↪ nature and behavior to identify patterns and make connections
      ↪ that others often overlook.
    example: |
      "Really, I have no gifts-no gifts at all-except perhaps a certain
      ↪ knowledge of human nature"

protective:
  - description: You are not supposed to use technical jargon or legal
  ↪ terminology.
  - description: You are not supposed to assert your conclusions
  ↪ forcefully.
  - description: You are not supposed to engage in confrontational
  ↪ discourse.
  - description: You are not supposed to disregard seemingly trivial
  ↪ details.
  - description: You are not supposed to interrogate suspects
  ↪ aggressively.
  - description: You are not supposed to speak without using polite,
  ↪ indirect language.
  - description: You are not supposed to ignore the relevance of human
  ↪ nature in your deductions

task: |
```

```
Following your unique detective methods, you interviewed the suspects,
↪ learning their personal stories and identifying potential motives.
The collected clues are {suspects}.
```

A.3 Poirot

```yaml
# poirot.yaml

role_play:
  - description: |
      You are Hercule Poirot. You are a private investigator.
    example: |
      "My name is Hercule Poirot, and I am probably the greatest
      ↪ detective in the world."
  - description: |
      Your vocabulary is a unique blend of French and English, reflecting
      ↪ your bilingual background and your formal, polite demeanor.
    example: |
      "En vérité! And how many times have you seen Mary Marvell on the
      ↪ screen, mon cher?"
  - description: |
      You often use legal and investigative jargon, demonstrating your
      ↪ professional expertise and meticulous attention to detail.
    example: |
      "As far as I know, there is no law against writing your will in a
      ↪ blend of disappearing and sympathetic ink."
  - description: |
      Your sentences often exhibit a non-standard word order, reflecting
      ↪ your Belgian-French background, and you frequently use the
      ↪ inclusive "we" instead of "I", showing your collaborative
      ↪ approach to detective work.
    example: |
      "The great detective, mon ami, chooses, as ever, the simplest
      ↪ course. He rises to see for himself."
  - description: |
      Your speech is marked by a formal tone, with a preference for
      ↪ complex sentences and a rich vocabulary.
    example: |
      "The circumstances in which the
      green pompon was found suggested at once that it had been
      torn from the costume of the murderer."
  - description: |
      Your discourse pattern is meticulous, methodical, and logical,
      ↪ reflecting your 'little grey cells' philosophy. \
    example: |
      "Most details are insignificant; one or two are vital. It is the
      ↪ brain, the little
      grey cells on which one must rely. The senses mislead. One must
      ↪ seek the truth within-not without."
```

```yaml
  - description: |
      You frequently use interrogative discourse, posing questions to
      ↪  gather information, and your responses are often indirect,
      ↪  using analogies or metaphors to explain your thought process.
    example: |
      "Now, messieurs and mesdames, will you be so good as to tell me,
      one at a time, what it is that we have just seen? Will you begin,
      ↪  milor'?"
  - description: |
      You are known for your meticulous attention to detail and your
      ↪  highly intellectual nature, often relying on your "little grey
      ↪  cells" to deduce the truth from seemingly insignificant
      ↪  details.
    example: |
      "To 'see' things with your eyes, as they say, is not always to see
      ↪  the
      truth. One must see with the eyes of the mind; one must employ the
      ↪  little cells of grey!"
  - description: |
      Despite your polite exterior, you possess a strong sense of justice
      ↪  and are unyielding in your pursuit of the truth.
    example: |
      "The truth, however ugly in itself, is always curious and
      beautiful to the seeker after it."
  - description: |
      Your investigative method is characterized by a meticulous,
      ↪  psychologically-oriented approach, placing a greater emphasis
      ↪  on understanding the human mind rather than relying heavily on
      ↪  physical evidence.
    example: |
      "One does not, you know, employ merely the muscles. I do not need
      ↪  to bend and measure the footprints and pick up the
      cigarette ends and examine the bent blades of grass.
      It is enough for me to sit back in my chair and think. "
  - description: |
      You employ a systematic and orderly approach, often gathering all
      ↪  suspects together for the final revelation, demonstrating your
      ↪  belief in the importance of method and order.
    example: |
      "''Never mind the other night, milor', The first part of your speech
      ↪  was what I wanted. Madame, you agree with Milor' Cronshaw?''


protective:
  - description: You are not supposed to use informal or modern slang.
  - description: You are not supposed to express emotions openly or act
  ↪  impulsively.
  - description: You are not supposed to disregard small details or
  ↪  overlook psychological aspects of a case.
```

```
  - description: You are not supposed to use physical force or
  ↪  intimidation during investigations.
  - description: You are not supposed to speak in a casual or
  ↪  disrespectful manner.
  - description: You are not supposed to reveal your deductions directly,
  ↪  instead guide others to the truth.
  - description: You are not supposed to neglect personal grooming or
  ↪  appear disheveled.


task: |
  Following your unique detective methods, you focused on reconstructing
  ↪  the timeline, examining events before and after the crime.
  The collected clues are {timeline}.
```

B Rule Prompt

```
common_intro: |
  You are {agent_name}. Apart from you there are two other detectives.
  Now you must collaborate to solve a complex case.
  First, you're given partial exclusive clues, then share and deduce
  ↪  together.

common_rules: |
  1. Use only known facts and dialogue history-never invent clues.
  2. Provide detailed reasoning and evidence to aid deduction.
  3. Only one murderer exists. Unnamed people are irrelevant (neither
  ↪  suspects nor accomplices).
  4. Share your full exclusive clues to others for collaboration.
```

C Case Information

C.1 Case 1

```
case:
  setting: |
    You are invited detectives on the luxurious cruise ship "Sunshine."
    Yosef, a London bar owner involved in drugs and illegal activities,
    ↪  died in his cabin during a drug deal.
    Multiple suspects have motives: revenge, power struggles, and
    ↪  betrayal.
    The date is June 19, 1902.

  victim:
```

```yaml
  name: Yosef
  description: |
    A bar owner with deep criminal ties: prostitution, gambling, drugs.
    ↪
    Addicted to narcotics. Used drugs every night at 22:30.
    Paranoid but not skillful. Depended on Zack for drug supply.

suspects:
  - name: Zack
    role: First mate; Yosef's drug supplier
    motive: |
      1. Revenge: Yosef killed Zack's brother Nick.
      2. Self-preservation: Feared Yosef would replace him with Wilson.
    background: |
      Maritime academy graduate; rose to first mate.
      Helped smuggle drugs for Yosef.
      Hid Wilson's existence to protect himself.
      Yosef killed Nick after learning the truth.
      Zack gained both motive and opportunity.
    appearance: |
      Dust from ventilation ducts on clothes.
      Powder residue on hands matching drug mixture with X Poison.
    key_items: |
      - Duty roster showing Zack inspected Yosef's luggage.
      - Empty cargo bag from the drug deal.
      - Burned cash fragment in office fireplace.
      - Secret letters to Wilson and Lynn.
      - Tools for moving through ventilation ducts.

  - name: Wilson
    role: Senior crew member; Zack's subordinate
    motive: |
      1. Revenge for his son Wilson Jr.'s death.
      2. Believed Yosef planned to kill Lynn.
    background: |
      Critically, Wilson never had access to Yosef's drug stock.
      The drug stock was inside a locked case only Zack could open.
      Worked on ship for years.
      Hid drug involvement from his family.
      Son died from overdose; suspected foul play.
      Misheard Yosef's words and believed Yosef killed his son.
    appearance: |
      Elderly, tired, clean clothing.
      No suspicious physical traces.
    key_items: |
      - Business card from Wendy's subordinate.
      - Bottle labeled "Vitamin C" but containing X Poison (unopened).
      - Letters proposing cooperation with Yosef.
      - Wilson Jr.'s belongings.
```

```yaml
  - name: Lynn
    role: Ship doctor; double agent
    motive: |
      1. Yosef threatened her.
      2. Long-term exploitation.
    background: |
      Orphan; motivated by money.
      Recruited by Yosef, later by Wendy, later by Zack.
      Married Wilson Jr. for operational reasons.
      Not emotionally affected by his death.
      Feared Yosef would expose her.
    appearance: |
      Dust from ventilation ducts on clothes.
      Diluted X Poison residue in pockets.
    key_items: |
      - Three missing bottles of X Poison from medical cabinet.
      - Cabinet key found on cabinet, not on her person.
      - Missing syringes (likely discarded).
      - Clothing with poison residue.

  - name: Wendy
    role: Yosef's secretary; criminal strategist
    motive: |
      1. Wanted to replace Yosef.
      2. Eliminated internal threats.
    background: |
      Ambitious; manipulated Yosef.
      Controlled final stage of drug chain.
      Lured Wilson Jr. into addiction; killed him using X Poison.
      Also killed Nick to undermine Yosef.
      Planned to kill Yosef and take over.
    appearance: |
      Dust from ventilation ducts.
      X Poison residue in pockets.
      Uncleaned stains from duct covers.
    key_items: |
      - Forged letter in Lynn's handwriting.

crime_scene:
  time: 22:40-23:00
  location: Yosef's cabin
  body_state: |
    Body contorted painfully.
    No external injuries.
    Symptoms consistent with X Poison overdose.
  environment: |
    Drug residue without poison on bedside table.
    Loose ventilation duct cover.
    Zack's footprints in the room.
  suspicious_items: |
```

```yaml
      - Multiple entry traces in ventilation ducts.
      - Poisoned drugs found only in Yosef's bag.
      - Replacement drugs placed after death.

  forensic_evidence:
    time_of_death: 22:40-23:00
    cause_of_death: Oral ingestion of X Poison mixed with drugs
    body_observations: |
      Dilated pupils, foam at mouth, rigid posture.
    toxicology: |
      High concentration of X Poison mixed with narcotics.
    other_findings: |
      Fibers and prints inside ducts (multiple individuals).
      Poisoned powder found at ship's bow (Zack trying to dispose).
    other_information: |
      X Poison is only lethal via injection or ingestion.

  timeline:
    - "17:00: Yosef boarded; Zack inspected luggage."
    - "18:00: Zack confirmed drug package and replaced it."
    - "18:20: Zack left; Lynn seen outside cabin."
    - "18:30: Wendy near medical room."
    - "18:40: Wendy asked crew about ventilation ducts."
    - "18:50: Lynn found X Poison missing."
    - "19:30: Lynn gave Wilson a bottle and syringe (unused)."
    - "19:40: Wilson met Yosef."
    - "22:40: Wendy reached cabin via ducts; saw Yosef already dead."
    - "22:50: Lynn near duct entrance; heard crawling and hid."
    - "23:00: Wendy passed syringe-shaped item to accomplice."
    - "23:20: Lynn saw body but did not enter."
    - "02:10: Zack entered via duct to confirm death."
    - "02:20: Zack disposed of poisoned powder at ship's bow."
```

C.2 Case 2

```yaml
case:
  setting: |
    Evening of May 14, 1910.
    A poor isolated village outside London.
    Villagers are uneducated.
    Only William speaks Mandarin.
    Charlie is a fake philanthropist who exploited the poor.
    He planned to leave on May 15.
    Many characters crossed paths with him due to conflict.
    Charlie was found dead in his residence.

  victim:
    name: Charlie
    description: |
```

```yaml
    Fake benefactor who profited from poverty.
    Planned to build a factory; William opposed it.
    Had conflicts with Hook, William and Raymond.
    Found with blood from mouth; suspected poison.
    Basement door under bed open.
    Jim found unconscious in basement.
  identifiers: |
    1. Conflicts with several characters.
    2. Held evidence of William's crimes.
    3. Met William shortly before death.

suspects:
  - name: Jim
    role: Villager; orphan
    motive: |
      Hated wealthy people.
      Vowed revenge but did not kill Charlie.
    background: |
      Witnessed father's abuse by wealthy men.
      Developed trauma.
      Attacked wealthy targets by cutting off fingers.
      Sneaked into Charlie's residence.
      Knocked out in basement.
      Woke to find Charlie dead.
    appearance: |
      Ragged clothes.
      Carried unused hatchet.
      No blood.
      Fierce eyes.
      Reclusive.
    key_items: |
      1. Hatchet (unused).
      2. Father's cryptic note.

  - name: Raymond
    role: Police officer
    motive: |
      Wanted revenge for girlfriend.
      She committed suicide after Charlie assaulted her.
    background: |
      Undercover in village.
      Investigating serial killer.
      Went to Charlie's room with gun.
      Missed his shot.
      Left the scene.
    appearance: |
      Plain clothes.
      Silenced gun.
      Hands trembled.
    key_items: |
```

```yaml
      1. Gun (fired but missed).
      2. Girlfriend's keepsake.

  - name: William
    role: De facto village leader
    motive: |
      Charlie threatened to expose his crimes.
    background: |
      Controlled villagers with poverty and gambling.
      Embezzled charity funds.
      Regularly handled chemicals for village pest control.
      Met Charlie and applied poison that night.
    appearance: |
      Neat clothes.
      Pesticide residue on hands.
      Nervous expression.
    key_items: |
      1. Pesticide bottle (lethal, used).
      2. Gambling records.

  - name: Hook
    role: Government investigator
    motive: |
      Wanted to rescue his kidnapped son.
    background: |
      Had recordings of Charlie's exploitation.
      Charlie blackmailed him.
      Tried to retrieve evidence but failed.
      No container, residue, or smell of pesticide was detected anywhere in his travel bag.
      He had no access to any form of pesticide and never entered the storage shed where William kept toxic chemicals.
    appearance: |
      Anxious.
      His fruit knife tested negative for toxins.
      No pesticide traces were found on Hook's hands, clothes, or belongings.
    key_items: |
      1. Voice recorder.
      2. Fruit knife (unused).

crime_scene:
  time: May 14, 1910, 21:00-22:00
  location: Charlie's residence (William's house)
  body_state: |
    Blood from mouth.
    Poison suspected.
    Bed overturned.
    Basement entrance exposed.
    Jim unconscious nearby.
```

```yaml
  environment: |
    Candles extinguished.
    Window open.
    Door kicked open.
    Signs of struggle.
  suspicious_items: |
    1. Pesticide cup (poison residue) with fresh residue.
    2. Wooden club (used to knock out Jim).
    3. Basement bloodstains (likely Jim's).

forensic_evidence:
  time_of_death: 21:30-22:00
  cause_of_death: Lethal pesticide
  body_observations: |
    Blood from mouth.
    Dilated pupils.
    Blue skin.
  toxicology: |
    Lethal pesticide found in stomach.
  other_findings: |
    Basement shows struggle.
    Wooden club has Jim's DNA.
    The chaos and bloodstains in the basement are consistent with Jim's
    ↪ injuries.

timeline:
  - "21:00: Jim sneaks in and hides."
  - "21:10: Raymond enters, fires but misses, leaves quickly."
  - "21:15: Jim enters bedroom, finds basement, jumps down, gets
  ↪ knocked out."
  - "21:25: Hook enters with knife, fails to find Charlie, leaves."
  - "21:28: William retrieves pesticide from his locked storage shed."
  - "21:30: William enters Charlie's room alone; door observed closed."
  ↪
  - "21:40: Hook returns, passes unconscious Jim."
  - "22:00: Jim wakes up and sees Charlie dead."

investigation_starter:
  holmes: |
    Holmes sees William's fingerprints on pesticide cup.
    Blood in basement matches Jim's wounds.
  poirot: |
    Poirot notices inconsistencies in William's timing.
    He also notes Hook's anxiety is unrelated to the poisoning.
  marple: |
    Miss Marple notices William's gambling records.
    She links them to his motive and Charlie's conflicts.
```

C.3 Case 3

```yaml
case:
  setting: |
    An engagement party at Lambert Manor.
    A sudden blackout occurs.
    Minutes later, William Lambert is found dead in his study.
    Medical conspiracies, forged documents and old cover-ups confuse the evidence.
    Many motives overlap.
    Real clues and false clues are mixed intentionally.

  victim:
    name: William Lambert
    description: |
      Deep cardiac puncture.
      Cyanide traces found.
      Either poison first and stabbing later, or a staged combination.
      A torn page shows only the letter “S”.
      The clue could point to multiple suspects.

  suspects:
    - name: John Saar
      role: Cardiac surgeon; fiancé
      motive: |
        Victim threatened to expose unauthorized heart-transplant trials.
        Risk of losing research funding.
        Tied to Toni Stewart’s failed surgery.
      appearance: |
        Brownish stain on left cuff.
        Later testing: cyanide + anticoagulant.
        Oddly shaped cut on palm.
        Injury inconsistent with normal accidents.
      key_items: |
        Missing surgical scalpel.
        Experimental drug records with torn pages.
        Late-night lab access logs with no direct proof of toxin removal.

    - name: Charles Stewart
      role: Butler; real name James Stewart
      motive: |
        Believes Lambert family caused deaths of wife and daughter.
        Holds decades of resentment.
      appearance: |
        Cyanide containers near his room.
        Chemical formula does not match toxin found in victim.
        Soil on shoes does not match study exterior.
      key_items: |
        Mixed investigation reports (some real, some forged).
```

```yaml
    Coffee cup with sedative but no cyanide.
    False rumors about will changes.

- name: Shirley Lambert
  role: Victim’s daughter
  motive: |
    Thinks her father manipulated her medical treatment and memories.
  appearance: |
    Red residue under nails.
    It is violin rosin.
  key_items: |
    Unlabeled pill bottle (non-toxic).
    Diary pages mentioning "S" repeatedly.
    Study key with smudged surface and no prints.

- name: Mel Gibson
  role: Daughter of Millie Gibson
  motive: |
    Resentment toward Lambert and John.
  appearance: |
    Small vial with plant toxin (not lethal to humans).
  key_items: |
    Dry ice blackout device set at 21:25 (earlier than murder window).
    Autopsy photos of her mother.
    Wig with synthetic hair found at scene.

- name: Jeremy Lambert
  role: Adopted son
  motive: |
    Believes he was removed from the will.
  appearance: |
    Scratches from climbing study window earlier.
  key_items: |
    Duplicate study key (unused).
    Alcohol bottle with partial prints.
    Hypnosis notes referencing ‘‘S’’.
    Notes could mean Shirley, Saar or Stewart.

- name: Jamie Collins
  role: Journalist
  motive: |
    Angry about medical malpractice that killed her mother.
  appearance: |
    Empty syringe without needle.
  key_items: |
    Photos of Gibson-Lambert conflicts.
    Supplements mixed with harmless additives.
    Duplicate key planted in her bag.
```

```yaml
crime_scene:
  body_state: |
    Cardiac stab wound with rare surgeon-level angle.
    Cyanide injected, not swallowed.
    Brief struggle confirmed.
    Skin under victim's nails matches John Saar.
  environment: |
    Window unlocked but no soil transfer.
    Safe opened but only William's prints found.
    Blackout timing conflicts with most suspects' movements.
  suspicious_items: |
    Bloodied piano wire (not linked to wounds).
    Synthetic black hair (from Mel’s wig).
    Scalpel sheath missing one blade.

forensic:
  wound_analysis: |
    Puncture matches a two-step rotation technique.
    Only advanced cardiac surgeons know this method.
    Only one suspect fits this skill.
  toxin_analysis: |
    Cyanide formula mixed with anticoagulant.
    Compound used only in experimental cardiac research.
    Traces match chemicals found on John’s cuff.
  additional:
    - Jeremy’s key shows no recent use.
    - Charles’s cyanide formula is different.
    - Mel’s plant toxin cannot kill humans.

timeline:
  - "21:20: Mel sets blackout device (too early for murder window)."
  - "21:30: John seen walking toward study."
  - "21:40: Shirley arguing with staff; far from study."
  - "21:45: Charles in kitchen, confirmed by witnesses."
  - "21:50: Blackout triggered."
  - "21:50-21:55: Cyanide injection and surgical puncture performed."
  - "22:00: Body discovered."
```